



Research paper

From brittle fracture to elastoplastic fatigue: An open-source phase field framework in Julia

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ABSTRACT

Phase-field methods are now widely adopted in fracture and fatigue modelling, and the associated variational formulations are well established in the literature. In computational practice and teaching, it is nevertheless useful to move between brittle fracture, elastic fatigue, ductile fracture, and elastoplastic fatigue within a single open-source implementation, while keeping a transparent link between the weak-form level, quadrature-point constitutive updates, and the outer staggered solution workflow, so as to lower implementation overhead and support reproducible studies. This paper presents an open-source Julia library built on the Gridap finite-element toolbox that addresses this need. The four regimes are integrated in one solver architecture and share AT1 and AT2 regularisations, Amor and spectral elastic energy decompositions, J_2 plasticity with isotropic and combined hardening, threshold-based fatigue degradation, and the fixed cycle-jump extrapolation strategy adopted here. A layered modular design with staggered minimisation organises the equilibrium problem, the phase-field problem, irreversibility and history-variable updates, and cycle-wise fatigue advancement under a common weak-form interface and shared data management, so that governing equations remain close to their code realisation. Eight benchmark problems are reported, from classical single-edge-notched tension and shear and double-notched ductile fracture to three-point bending elastic fatigue and elevated-temperature compact-tension elastoplastic fatigue in a GH4169 nickel-base superalloy. The predicted force–displacement curves and crack-growth histories agree well with published numerical references and experimental data. Representative studies on mesh refinement and on the cycle-jump step size illustrate how these numerical choices affect key response quantities and wall-clock cost. The source code, reproducible package environment, and self-contained benchmark scripts are openly available to support further work and cross-regime comparison.

1. Introduction

Phase field fracture has, over the past two decades, become one of the most influential frameworks in computational fracture mechanics. Its central idea originates from the variational fracture theory of Francfort and Marigo [1], which reformulated the classical Griffith criterion as the minimisation of a free-discontinuity energy functional and removed the need to prescribe crack trajectories a priori. Bourdin, Francfort, and Marigo [2] subsequently introduced a regularised approximation in which the geometrically sharp crack surface is replaced by a continuous scalar damage field distributed over a diffuse band of width proportional to the regularisation length l_0 . Within this setting, crack nucleation, propagation, branching, and coalescence emerge naturally from energy minimisation, without any geometric tracking or remeshing. The rigorous Γ -convergence foundations were established in [3]. Comprehensive reviews by Wu et al. [4] and Ambati et al. [5] document the broad engineering applications that have followed.

The physical scope of the framework was subsequently extended in a systematic way. In the elastic regime, Miehe, Welschinger, and Hofacker [6] established a thermodynamically consistent staggered scheme with spectral energy decomposition, ensuring that only tensile strains drive damage and suppressing spurious cracking under compression. Amor, Marigo, and Maurini [7] proposed the volumetric–deviatoric split as a computationally attractive alternative that avoids spectral decomposition while retaining accurate crack-path prediction in tension- and bending-dominated problems. Pham et al. [8] introduced the AT1 regularisation, which incorporates a finite elastic threshold, providing a clearer physical interpretation in brittle fracture settings. More recent contributions have continued to strengthen the variational and numerical foundations of the method through adaptive active-set strategies and predictor–corrector mesh adaptivity [9]. Gong et al. [10], for example, recently validated phase field predictions of

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Nomenclature

$\Omega, \partial\Omega$	body domain and its boundary	d	phase field variable (0: intact; 1: broken)
$\partial\Omega_D, \partial\Omega_N$	Dirichlet/Neumann boundaries	l_0	phase field regularisation length
n_d	spatial dimension (2 or 3)	G_c	critical energy release rate
$\mathbf{u}, \bar{\mathbf{u}}$	displacement/prescribed displacement	$\gamma(d, \nabla d)$	crack surface density function
$\mathbf{t}, \mathbf{b}$	surface traction/body force	$g(d)$	degradation function
$\mathbf{n}$	outward unit normal	η_k	residual stiffness parameter
$\boldsymbol{\epsilon}, \boldsymbol{\epsilon}^e, \boldsymbol{\epsilon}^p$	total/elastic/plastic strain	$\psi^e, \psi^{e\pm}$	elastic energy density (tensile/compressive)
$\mathbb{C}$	fourth-order elasticity tensor	K_d	Amor split modulus ($\lambda + 2G/n_d$)
E, ν	Young's modulus/Poisson's ratio	$\mathbf{I}, \mathbb{I}^{\text{sym}}$	second-/fourth-order identity tensors
λ, G	Lamé parameters	δ_{ij}	Kronecker delta
$\bar{\boldsymbol{\sigma}}$	effective (undamaged) stress	$\mathcal{H}$	crack-driving history variable
$\boldsymbol{\sigma}$	nominal (macroscopic) stress	β	Taylor–Quinney weight
$\bar{\sigma}_{\text{eq}}$	von Mises equivalent stress	$\bar{\alpha}$	instantaneous fatigue driving force
$\tilde{s}_{ij}$	deviatoric stress components	$\bar{\alpha}$	cumulative fatigue history variable
F	yield function	$f(\bar{\alpha})$	fatigue degradation function
$\dot{\gamma}, \Delta\gamma$	plastic multiplier/increment	α_T	fatigue threshold
$\mathbf{N}$	unit flow direction	ΔN	cycle-jump size
α_p	equivalent plastic strain	N_{sub}	substeps per load cycle
$\kappa(\alpha_p)$	isotropic hardening function	σ_{y0}	initial yield stress
n	hardening exponent	H	linear hardening modulus
ψ^p	plastic stored energy density		

fracture toughness in octet-truss lattices against experiments, illustrating the method's expanding applicability to architected lightweight materials.

In the fatigue regime, Carrara et al. [11] and Schreiber et al. [12] independently introduced cumulative fatigue variables to drive the progressive degradation of fracture toughness under cyclic loading, extending the phase field framework to fatigue crack growth. Cycle-jump strategies developed by Seiler et al. [13], Loew et al. [14], and Kristensen et al. [15] reduced the cost of high-cycle simulations by one to two orders of magnitude, enabling engineering-scale fatigue prediction. These advances have been applied successfully to hydrogen-assisted cracking and fatigue, non-isothermal fatigue, shape-memory-alloy fatigue, and fatigue in porous media [16–21]. Coupling phase field fracture with J_2 plasticity further pushed the framework towards ductile metallic failure. Ambati et al. [22] and Borden et al. [23] established the theoretical basis of ductile phase field fracture and the paradigm in which plastic dissipation contributes to the crack driving force. Subsequent studies extended this line towards thermo-elasto-plastic, porous-ductile, and more general mixed-mode formulations [24–26]. Khalil et al. [27] subsequently developed a complete elastoplastic fatigue phase field formulation in the effective stress space; Xie et al. [28] validated the approach against high-temperature fatigue experiments on GH4169 compact-tension specimens; and Li et al. [29] investigated cyclic-plasticity–damage coupling and developed an adaptive cycle-jump strategy. Together, these developments form a coherent chain from brittle elastic fracture to elastic fatigue, ductile fracture, and elastoplastic fatigue, each step supported by both rigorous theory and experimental validation.

In parallel with these regime-specific extensions, adaptive and multiscale phase-field strategies have also been developed to improve computational efficiency and to handle heterogeneous or microstructured materials [30,31]. These directions are largely orthogonal to the regime-unification framework adopted in the present work.

Despite this theoretical maturity, practical adoption remains comparatively limited, partly because the implementation effort can still be considerable. In practice, several representative implementation routes have been developed for phase-field fracture and fatigue modelling, relying on different finite-element ecosystems and programming languages. Table 1 summarises a selection of representative open or

publicly described implementations together with their typical characteristics, illustrating how different routes differ in ecosystem dependence, modelling transparency, and the physical regimes that each work primarily addresses.

These works have considerably lowered the entry barrier for phase-field fracture modelling, while each implementation typically focuses on a specific software ecosystem, material system, or physical regime. Earlier and contemporary open-source efforts that follow these routes include Abaqus subroutine implementations [38–40], FEniCS-based formulations [41–43], and MOOSE-based developments [44]. For instance, several publicly described implementations already cover brittle fracture or elastic fatigue, but unified support for elastoplastic fatigue together with cycle-jump acceleration remains comparatively limited; in addition, comparisons across implementation routes within a single codebase are not straightforward. Against this background, further developing an open-source implementation that connects brittle fracture, elastic fatigue, ductile fracture, and elastoplastic fatigue within the same code framework remains useful for reproducible studies and for cross-comparison among methods.

The present work addresses this gap. Julia is chosen mainly because its mathematical syntax allows variational equations to be expressed closely to their analytical form. In addition, Julia provides a single-language environment: modifications at the quadrature-point level, constitutive updates, and global assembly all take place in the same language, which reduces cross-language overhead. The finite-element backbone is provided by Gridap [45,46], whose weak-form API allows the bilinear forms of the displacement and phase-field subproblems to be written directly from the governing equations. Quadrature-point plastic, crack-history, and fatigue-history variables are managed through Gridap's cell-array infrastructure, so the return-mapping algorithm integrates with the global assembly loop. Meshes are generated parametrically through Gmsh [47] and GridapGmsh, and a lightweight graphical interface based on GLMakie.jl supports real-time visualisation of damage evolution and load–response curves. The interface is intended primarily for teaching and rapid case inspection rather than as a general finite-element workbench; representative screenshots are provided in Appendix B, and Linux is the recommended platform for routine use.

Table 1
Representative implementation routes for phase-field fracture and fatigue models.

Implementation route	Representative work	Implementation path	Typical characteristics	Coverage relative to the present scope
Abaqus UEL/UMAT	[32]	Implements displacement and phase-field elements through Abaqus UEL/UMAT user subroutines	Built on a commercial finite-element platform with mature pre/post-processing and nonlinear solvers	Addresses elastoplastic phase-field fracture, but does not provide an open-source unified codebase
COMSOL PDE module	[33,34]	Defines the weak forms directly within the COMSOL Multiphysics PDE module, with extensions to specialised material systems such as quasi-crystals	Flexible PDE-level prototyping with graphical model construction and multiphysics coupling	Focuses on brittle crack propagation in selected material systems; elastoplastic fatigue is not considered
FEniCS/FEniCSx	[35]	Provides a Python high-level interface for phase-field fracture and fatigue based on FEniCSx	Concise weak-form interface and an actively maintained open-source ecosystem	Covers brittle fracture and elastic fatigue; unified elastoplastic fatigue with cycle-jump acceleration is not its main focus
MOOSE	[36]	Builds on the MOOSE framework to model phase-field fracture in functionally graded materials	Mature multiphysics infrastructure with established parallel-computing capability	Targeted at brittle fracture in specific material systems; the full fracture-fatigue chain is not covered
Julia/Gridap	[37]	Implements a brittle phase-field fracture model in Julia using the Gridap finite-element toolbox	Single-language environment with a clear correspondence between weak forms and code	Concentrates on brittle-fracture examples; elastoplasticity and fatigue are not addressed

The library covers elastic fracture, elastic fatigue, ductile fracture, and elastoplastic fatigue within a single layered architecture, and supports AT1/AT2 regularisation, Amor and spectral energy decompositions, effective-stress J_2 plasticity with power-law, linear, and perfect-plastic hardening, Taylor–Quinney weighted crack driving, threshold-based fatigue degradation, and cycle-jump acceleration. Eight benchmark problems are used to assess correctness, ranging from classical single-edge-notched tension and shear to ductile double-notched specimens, three-point-bending fatigue, and high-temperature GH4169 compact-tension fatigue. The predicted load–displacement curves and crack-growth histories agree well with published numerical and experimental results, and the complete source code is openly available.

The remainder of the paper is organised as follows. Section 2 establishes the unified mathematical framework covering all four physical regimes. Section 3 presents the software architecture, constitutive implementation, and staggered solution strategy. Section 4 reports the benchmark validations. Section 5 discusses computational performance. Section 6 concludes the paper. Appendix A summarises the code structure and usage guide, and Appendix B presents representative screenshots of the demonstration GUI.

2. Mathematical framework

This chapter establishes the unified variational framework that underpins the four physical regimes considered in the library. Section 2.1 first introduces the complete energy functional and the weak-form governing equations, which form the basis for the rest of the chapter. The subsequent sections then introduce, in turn, the phase-field regularisation, the energy decomposition, the elastoplastic constitutive behaviour, and the fatigue degradation. This order mirrors the layered software architecture: the solver operates on the same weak-form interface, and additional physical complexity is introduced by activating extra constitutive ingredients rather than by changing the solver structure.

2.1. Unified variational framework and governing equations

Consider a solid occupying the domain $\Omega \subset \mathbb{R}^{n_d}$ ($n_d = 2$ or 3), with boundary $\partial\Omega = \partial\Omega_D \cup \partial\Omega_N$. Dirichlet conditions $\mathbf{u} = \bar{\mathbf{u}}$ are prescribed on $\partial\Omega_D$, while tractions $\mathbf{t}$ act on $\partial\Omega_N$. Within the small-strain setting, the total strain tensor is defined as the symmetric gradient of the displacement field:

$$\boldsymbol{\varepsilon} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T) \quad (1)$$

and is additively decomposed into elastic and plastic parts:

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p \quad (2)$$

For a linear isotropic elastic material, the fourth-order elasticity tensor is

$$\mathbb{C} = \lambda \mathbf{I} \otimes \mathbf{I} + 2G \mathbb{I}^{\text{sym}} \quad (3)$$

where $\lambda = E\nu/[(1+\nu)(1-2\nu)]$ and $G = E/[2(1+\nu)]$ are the Lamé parameters. The effective (undamaged) stress is defined as

$$\tilde{\boldsymbol{\sigma}} = \mathbb{C} : \boldsymbol{\varepsilon}^e = \lambda \text{tr}(\boldsymbol{\varepsilon}^e) \mathbf{I} + 2G \boldsymbol{\varepsilon}^e \quad (4)$$

and the elastic strain energy density reads

$$\psi^e = \frac{1}{2} \boldsymbol{\varepsilon}^e : \mathbb{C} : \boldsymbol{\varepsilon}^e = \frac{\lambda}{2} [\text{tr}(\boldsymbol{\varepsilon}^e)]^2 + G \boldsymbol{\varepsilon}^e : \boldsymbol{\varepsilon}^e \quad (5)$$

The regularised total potential energy of the system is written in the unified form

$$\begin{aligned} \Pi(\mathbf{u}, d) = & \int_{\Omega} \left[g(d) \psi^{e+}(\boldsymbol{\varepsilon}^e) + \psi^{e-}(\boldsymbol{\varepsilon}^e) + \beta \psi^p(\alpha_p) \right] d\Omega \\ & + \int_{\Omega} f(\bar{\alpha}) G_c \gamma(d, \nabla d) d\Omega \end{aligned} \quad (6)$$

where

$$g(d) = (1-d)^2 + \eta_k \quad (7)$$

is the degradation function, satisfying $g(0) \approx 1$, $g(1) \approx 0$, and $g'(1) = 0$, with the residual stiffness $\eta_k = 10^{-10}$ preventing ill-conditioning in fully damaged regions. In (6), $\psi^{e\pm}$ denote the tensile and compressive parts of the elastic strain energy, defined in Section 2.3; $\beta \psi^p$ is the Taylor–Quinney weighted plastic stored energy, discussed in Section 2.4.3; $f(\bar{\alpha})$ is the fatigue degradation function introduced in Section 2.5.1; and $\gamma(d, \nabla d)$ is the crack surface density functional defined in Section 2.2.

Taking the first variation of (6) with respect to the displacement field $\mathbf{u}$ and the phase field d yields two coupled subproblems. The strong form of equilibrium is

$$\nabla \cdot \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \quad \text{in } \Omega \quad (8)$$

with body force $\mathbf{b}$ set to zero hereafter. The nominal (macroscopic) stress is given by

$$\boldsymbol{\sigma} = g(d) \tilde{\boldsymbol{\sigma}} = g(d) \mathbb{C} : (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) \quad (9)$$

The degradation function acts only on the nominal stress entering equilibrium and does not appear explicitly in the yield function defined in

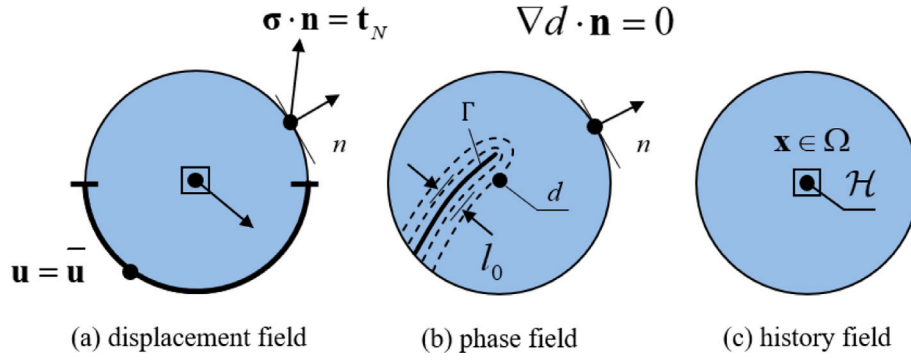


Fig. 1. Schematic of the three coupled subproblems in the phase field framework. (a) Displacement field $\mathbf{u}$: Dirichlet condition $u = u_D$ is prescribed on $\partial\Omega_D$; Neumann traction $\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{t}_N$ acts on $\partial\Omega_N$. (b) Phase field d : the diffuse crack Γ of regularisation width $\sim 2l_0$ evolves according to the phase field equation with natural boundary condition $\nabla d \cdot \mathbf{n} = 0$. (c) History field $\mathcal{H}$: stores the maximum tensile crack-driving energy density, ensuring irreversible crack growth pointwise.

the effective stress space. The weak form of the displacement problem is: for all $\mathbf{v} \in \mathcal{V}_0$,

$$\begin{aligned} \int_{\Omega} g(d) \mathbb{C} : \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega \\ = \int_{\partial\Omega_N} \mathbf{t} \cdot \mathbf{v} d\Gamma + \int_{\Omega} g(d) \mathbb{C} : \boldsymbol{\varepsilon}^p : \boldsymbol{\varepsilon}(\mathbf{v}) d\Omega \end{aligned} \quad (10)$$

Here the full elastic stiffness is degraded uniformly by $g(d)$, rather than applying the tension–compression split directly at the equilibrium level. The energy decomposition is used only to define the crack-driving history variable $\mathcal{H}$. This hybrid formulation preserves the linearity of the displacement problem for fixed d and $\boldsymbol{\varepsilon}^p$; the nonlinearity introduced by plasticity is treated by Picard iteration (Section 3.2). The corresponding phase field subproblem differs between regularisation models and will be stated in Section 2.2. Crack irreversibility is enforced through the point-wise constraint $d_{n+1} \geq d_n$ after each phase field solve.

Fig. 1 provides a schematic overview of the three coupled subproblems that constitute the variational framework: the displacement subproblem determines the deformed configuration for a given damage state; the phase field subproblem updates the diffuse crack variable d given the current crack-driving history; and the history-field update records the maximum tensile energy density and enforces irreversibility pointwise. The staggered solver cycles through these three subproblems at every load increment, as detailed in Section 3.2.

This unified framework covers all four physical regimes by selectively activating its constitutive ingredients, and this activation logic is also the organising principle of the software architecture. Elastic fracture corresponds to $\boldsymbol{\varepsilon}^p \equiv \mathbf{0}$, $\beta = 0$, and $f \equiv 1$, reducing the formulation to the standard phase field fracture setting with $\mathcal{H} = \max_{\tau \leq t} \psi^e$. Elastic fatigue activates the degradation function $f(\bar{\alpha})$ together with the cycle-jump scheme of Section 2.5.2. Ductile fracture activates return mapping while keeping $f \equiv 1$, so the crack-driving force gains the additional plastic contribution $\beta\psi^p$. Elastoplastic fatigue activates all ingredients simultaneously: return mapping, $\beta\psi^p$, $f(\bar{\alpha})$, and cycle-jump acceleration. Each subsequent subsection defines one constituent of this layered structure.

2.2. Phase field regularisation: AT1 and AT2 models

The crack surface density functional $\gamma(d, \nabla d)$ replaces the geometrically sharp crack surface energy $\int_{\Gamma} G_c d\Gamma$ by the volume integral $\int_{\Omega} G_c \gamma d\Omega$. The regularisation length $l_0 > 0$ controls the width of the diffuse crack band; as $l_0 \rightarrow 0$, the regularised functional converges to the sharp-crack energy in the Γ -convergence sense [3]. Two regularisation models are implemented in the present library.

The AT2 model adopts

$$\gamma_{AT2}(d, \nabla d) = \frac{d^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \quad (11)$$

Table 2

Comparison of AT1 and AT2 phase field models.

	AT2	AT1
$\gamma(d, \nabla d)$	$\frac{d^2}{2l_0} + \frac{l_0}{2} \nabla d ^2$	$\frac{3}{8} \left(\frac{d}{l_0} + l_0 \nabla d ^2 \right)$
Damage profile	Smooth (exponential)	Linear with compact support
Elastic threshold	None	$\sigma_c = \sqrt{3G_c E / (8l_0)}$

and yields the exponential one-dimensional damage profile $d(x) = \exp(-|x|/l_0)$. Damage may start to accumulate from the very first loading step, which makes AT2 especially suitable for modelling gradual degradation under fatigue loading.

The AT1 model adopts

$$\gamma_{AT1}(d, \nabla d) = \frac{3}{8} \left(\frac{d}{l_0} + l_0 |\nabla d|^2 \right) \quad (12)$$

and possesses a finite elastic threshold $\sigma_c = \sqrt{3G_c E / (8l_0)}$, below which no damage is initiated. This finite threshold gives AT1 a clearer physical interpretation in brittle fracture problems (see Table 2).

The physical effect of l_0 on the diffuse crack band is illustrated schematically in Fig. 2. The top row visualises the compact-support character associated with the AT1 regularisation, whereas the bottom row shows the exponential AT2 profile for the same set of regularisation lengths. In both models, decreasing l_0 progressively contracts the regularised damage zone from a broad diffuse band to a much sharper localisation, thereby approaching the geometrically sharp crack in the Γ -convergence limit. In engineering practice, l_0 is treated as a material length scale calibrated to the critical stress: for AT1, the relation $\sigma_c = \sqrt{3G_c E / (8l_0)}$ provides a direct physical calibration.

Based on these crack surface density functionals, the strong and weak forms of the phase field subproblem are as follows. For the AT2 model, the strong form reads

$$g'(d)\mathcal{H} + \frac{fG_c}{l_0} d - fG_c l_0 \Delta d = 0 \quad \text{in } \Omega \quad (13)$$

and the corresponding weak form is: for all $q \in \mathcal{V}_d$,

$$\begin{aligned} \int_{\Omega} \left[\left(\frac{fG_c}{l_0} + 2\mathcal{H} \right) dq + fG_c l_0 \nabla d \cdot \nabla q \right] d\Omega \\ = \int_{\Omega} 2\mathcal{H} q d\Omega \end{aligned} \quad (14)$$

For the AT1 model, the strong form becomes

$$g'(d)\mathcal{H} + \frac{3fG_c}{8l_0} d - \frac{3fG_c l_0}{4} \Delta d = 0 \quad \text{in } \Omega \quad (15)$$

and the weak form reads

$$\int_{\Omega} \left[2\mathcal{H} dq + \frac{3fG_c l_0}{4} \nabla d \cdot \nabla q \right] d\Omega$$

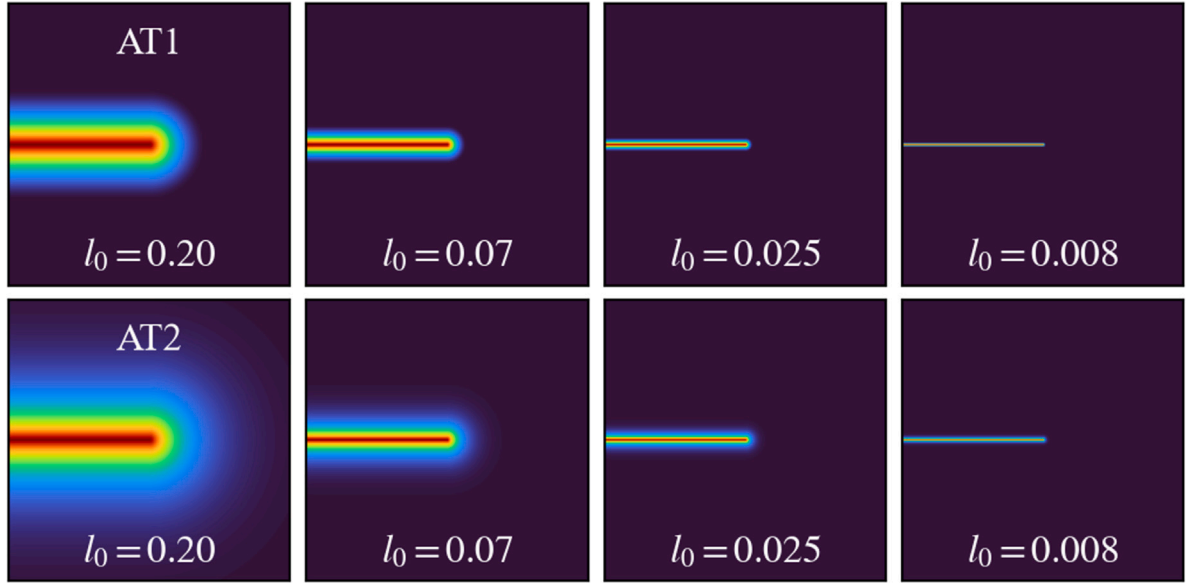


Fig. 2. Schematic comparison of the influence of the phase field regularisation length l_0 on the diffuse crack profile for the two regularisation models implemented in the library. The top row shows an AT1-style compact-support profile, and the bottom row shows the AT2 exponential profile $d = \exp(-r/l_0)$, for the same horizontal crack embedded in a square domain; here r denotes the distance from a point to the nearest location on the crack. In both rows, the diffuse damage band narrows from left to right as l_0 decreases, approaching the sharp-crack limit. White text inside the first-column panels identifies the AT1 and AT2 rows, and the numerical value of l_0 is shown at the bottom of each panel.

$$= \int_{\Omega} \left(2H - \frac{3fG_c}{8l_0} \right) q \, d\Omega \quad (16)$$

In both cases, the phase field equation is linear in d for prescribed H and f , which is central to the efficiency of the staggered solution strategy. The natural boundary condition $\nabla d \cdot \mathbf{n} = 0$ is imposed on the boundary.

2.3. Energy decomposition: Amor volumetric–deviatoric split and Miehe spectral split

Damage is driven only by the tensile part ψ^{e+} of the elastic energy, while the compressive part ψ^{e-} is excluded from the crack-driving force in order to prevent non-physical crack growth under compression. Two decomposition schemes are implemented and can be selected independently at runtime.

The Amor volumetric–deviatoric split separates the elastic energy into volumetric expansion and deviatoric distortion:

$$\psi_A^{e+} = \frac{K_d}{2} \langle \text{tr}(\epsilon^e) \rangle_+^2 + G \, \text{dev}(\epsilon^e) : \text{dev}(\epsilon^e), \quad (17)$$

$$\psi_A^{e-} = \frac{K_d}{2} \langle \text{tr}(\epsilon^e) \rangle_-^2$$

where $K_d = \lambda + 2G/n_d$ is the dimension-dependent modulus and $\langle \cdot \rangle_{\pm}$ denote the positive and negative Macaulay brackets. This split avoids eigenvalue calculations and is therefore computationally attractive for tension- and bending-dominated problems.

The Miehe spectral split is obtained from the spectral decomposition $\epsilon^e = \sum_a \epsilon_a^e \mathbf{n}_a \otimes \mathbf{n}_a$ of the elastic strain tensor:

$$\psi_S^{e+} = \frac{\lambda}{2} \langle \text{tr}(\epsilon^e) \rangle_+^2 + G \sum_a \langle \epsilon_a^e \rangle_+^2, \quad (18)$$

$$\psi_S^{e-} = \frac{\lambda}{2} \langle \text{tr}(\epsilon^e) \rangle_-^2 + G \sum_a \langle \epsilon_a^e \rangle_-^2$$

The spectral split provides more accurate crack-path prediction in shear-dominated problems and is indispensable for benchmarks such as the single-edge notched shear test; a quantitative comparison of the two decompositions is given in Section 4.1.2.

It is important to note that plane strain still requires a fully three-dimensional stress state. Under the plane-strain constraint $\epsilon_{33} = 0$,

plastic incompressibility implies $\dot{\epsilon}_{33}^p = -\dot{\epsilon}_{11}^p - \dot{\epsilon}_{22}^p \neq 0$ once plastic flow occurs, and therefore $\epsilon_{33}^e = -\epsilon_{33}^p \neq 0$. The out-of-plane stress component

$$\begin{aligned} \sigma_{33} &= \lambda \, \text{tr}(\epsilon^e) + 2G\epsilon_{33}^e \\ &= \lambda(\epsilon_{11}^e + \epsilon_{22}^e + \epsilon_{33}^e) + 2G\epsilon_{33}^e \end{aligned} \quad (19)$$

is therefore generally non-zero and must be retained in the yield evaluation, in the von Mises equivalent stress, and in the elastic energy density. Thus, although the boundary-value problem is solved on a two-dimensional domain, each quadrature point carries the complete three-dimensional stress and strain state.

2.4. Elastoplastic constitutive model

2.4.1. J_2 plasticity in the effective stress space

Following the effective-stress concept of continuum damage mechanics, the J_2 yield function is formulated entirely in the effective (undamaged) stress space and contains no explicit dependence on the phase field damage variable:

$$F(\tilde{\sigma}, \alpha_p) = \tilde{\sigma}_{\text{eq}} - \sqrt{\frac{2}{3}} \kappa(\alpha_p) \leq 0 \quad (20)$$

where the von Mises equivalent stress is computed from the full three-dimensional deviatoric stress:

$$\begin{aligned} \tilde{\sigma}_{\text{eq}} &= \sqrt{\frac{3}{2} \text{dev}(\tilde{\sigma}) : \text{dev}(\tilde{\sigma})} \\ &= \sqrt{\frac{3}{2} (\tilde{s}_{11}^2 + \tilde{s}_{22}^2 + \tilde{s}_{33}^2 + 2\tilde{s}_{12}^2)} \end{aligned} \quad (21)$$

The scalar $\alpha_p \geq 0$ denotes the equivalent plastic strain and $\kappa(\alpha_p)$ the isotropic hardening function. The central advantage of the effective-stress formulation is that damage reduces the structural load-carrying capacity through the nominal stress $\sigma = g(d)\tilde{\sigma}$, while the plastic flow rule acts on the undamaged material response and therefore does not artificially suppress plasticity in partially damaged regions.

The associative flow rule and the evolution of the equivalent plastic strain are

$$\dot{\epsilon}^p = \dot{\gamma} \mathbf{N}, \quad \mathbf{N} = \frac{3}{2} \frac{\text{dev}(\tilde{\sigma})}{\tilde{\sigma}_{\text{eq}}}, \quad \dot{\alpha}_p = \sqrt{\frac{2}{3}} \dot{\gamma} \quad (22)$$

Table 3
Isotropic hardening laws implemented in the library.

	Power-law	Linear	Perfect
$\kappa(\alpha_p)$	$\sigma_{y0} \left(1 + \frac{E}{\sigma_{y0}} \alpha_p\right)^n$	$\sigma_{y0} + H \alpha_p$	σ_{y0}
$\kappa'(\alpha_p)$	$nE \left(1 + \frac{E}{\sigma_{y0}} \alpha_p\right)^{n-1}$	H	0
$\psi^p(\alpha_p)$	$\frac{\sigma_{y0}^2}{E(n+1)} \left[\left(1 + \frac{E}{\sigma_{y0}} \alpha_p\right)^{n+1} - 1 \right]$	$\sigma_{y0} \alpha_p + \frac{1}{2} H \alpha_p^2$	$\sigma_{y0} \alpha_p$
Parameters	σ_{y0}, E, n	σ_{y0}, H	σ_{y0}

where $\dot{\gamma} \geq 0$ is the plastic multiplier. The flow direction $\mathbf{N}$ is trace-free and therefore enforces plastic incompressibility. The Karush–Kuhn–Tucker conditions

$$\dot{\gamma} \geq 0, \quad F \leq 0, \quad \dot{\gamma} F = 0 \quad (23)$$

together with the consistency condition close the rate-independent plasticity model.

2.4.2. Isotropic hardening laws

Three isotropic hardening functions $\kappa(\alpha_p)$ are implemented, together with their thermodynamically conjugate plastic stored energy densities $\psi^p(\alpha_p)$, summarised in Table 3. The plastic stored energy is the fraction of the plastic work retained in the material microstructure and is the direct source of the $\beta\psi^p$ contribution to the crack-driving force.

The power-law hardening model is written as

$$\kappa(\alpha_p) = \sigma_{y0} \left(1 + \frac{E}{\sigma_{y0}} \alpha_p\right)^n \quad (24)$$

Its derivative, required for the Jacobian in the return-mapping Newton iteration (Section 3.3), is

$$\kappa'(\alpha_p) = nE \left(1 + \frac{E}{\sigma_{y0}} \alpha_p\right)^{n-1} \quad (25)$$

and the associated plastic stored energy is

$$\psi^p(\alpha_p) = \frac{\sigma_{y0}^2}{E(n+1)} \left[\left(1 + \frac{E}{\sigma_{y0}} \alpha_p\right)^{n+1} - 1 \right] \quad (26)$$

Linear hardening, $\kappa(\alpha_p) = \sigma_{y0} + H \alpha_p$, is used when the material data are given in terms of a hardening modulus H , with corresponding stored energy $\psi^p = \sigma_{y0} \alpha_p + \frac{1}{2} H \alpha_p^2$. Perfect plasticity corresponds to $H = 0$ (or $n = 0$) and provides a conservative lower bound of the hardening response. The influence of hardening-law selection on ductile fracture and fatigue crack growth is examined in Sections 4.2.1 and 4.4.

The CT benchmark of Section 4.4.1 additionally employs a combined Voce isotropic and Armstrong–Frederick kinematic hardening model, calibrated directly from the experimental data of Xie et al. [28] for GH4169 at 550 °C. The kinematic hardening backstress $\mathbf{X}$ evolves according to $\dot{\mathbf{X}} = C\dot{\epsilon}^p - \gamma \mathbf{X} \dot{\alpha}_p$, and the yield function is evaluated in the translated stress space $\tilde{\boldsymbol{\sigma}} - \mathbf{X}$. This constitutive update is implemented as a direct reproduction of the reference model and is encapsulated in the PlasticFatigue_CT.jl script; it does not alter the shared solver infrastructure.

2.4.3. Crack-driving history variable

The crack-driving history variable combines the tensile elastic energy and the Taylor–Quinney weighted plastic stored energy, and is defined through the history maximum so as to enforce monotonicity of the crack-driving force and irreversibility of fracture:

$$H(\mathbf{x}, t) = \max_{\tau \leq t} [\psi^{e+}(\mathbf{x}, \tau) + \beta \psi^p(\mathbf{x}, \tau)] \quad (27)$$

The dimensionless weight $\beta \in [0, 1]$ represents the fraction of plastic work stored in the lattice. Classical Taylor–Quinney experiments [48] indicate that roughly 85%–95% of the plastic work is converted to heat, which motivates the physically calibrated range $\beta \approx 0.05$ –0.15; $\beta = 1$ serves as a conservative upper bound. For purely elastic problems, $\beta = 0$ and $\psi^p \equiv 0$, so that the history variable reduces to $\max_{\tau \leq t} \psi^{e+}$.

2.5. Fatigue degradation mechanism

2.5.1. Fatigue accumulation variable and degradation function

Under cyclic loading, the effective fracture toughness is progressively reduced through the fatigue degradation function $f(\bar{\alpha})$, so that the effective toughness becomes $f(\bar{\alpha})G_c$, with $0 < f \leq 1$. The instantaneous fatigue driving force is defined as

$$\bar{\alpha}(\mathbf{x}, t) = g(d) [\psi^{e+}(\mathbf{x}, t) + \beta \psi^p(\mathbf{x}, t)] \quad (28)$$

where the factor $g(d)$ ensures that fully damaged regions no longer accumulate additional fatigue damage. The cumulative fatigue history variable $\bar{\alpha}$ grows monotonically by accumulating only the positive increments of $\bar{\alpha}$:

$$\bar{\alpha}_{n+1} = \bar{\alpha}_n + \Delta \bar{\alpha}_n, \quad \Delta \bar{\alpha}_n = \max(0, \bar{\alpha}_{n+1} - \bar{\alpha}_n) \quad (29)$$

This accumulation rule captures the irreversibility of fatigue damage: only loading half-cycles that generate a new maximum in the driving force contribute to degradation, whereas unloading half-cycles do not.

The threshold-based fatigue degradation function is taken as

$$f(\bar{\alpha}) = \begin{cases} 1, & \bar{\alpha} \leq \alpha_T \\ \left(\frac{2\alpha_T}{\bar{\alpha} + \alpha_T} \right)^2, & \bar{\alpha} > \alpha_T \end{cases} \quad (30)$$

with the threshold $\alpha_T = G_c/(12l_0)$. When $\bar{\alpha}$ remains below the threshold, no degradation occurs; once the threshold is exceeded, f decreases monotonically towards zero. The function is continuous at the threshold and numerically robust, while also reflecting the endurance-limit behaviour commonly observed in metallic materials.

2.5.2. Cycle-jump acceleration scheme

For high-cycle fatigue problems ($N_{\text{total}} \sim 10^4$ – 10^6), cycle-by-cycle simulation is prohibitively expensive. The cycle-jump technique assumes that the fatigue state varies approximately linearly over a block of ΔN consecutive cycles. For each block, one representative cycle is simulated with N_{sub} substeps, and the per-cycle fatigue increment $\Delta \bar{\alpha}_{\text{cycle}} = \sum_i \Delta \bar{\alpha}_i$ is accumulated according to Eq. (29). This increment is then extrapolated to the whole cycle block:

$$\bar{\alpha} \leftarrow \bar{\alpha} + \Delta N \cdot \Delta \bar{\alpha}_{\text{cycle}} \quad (31)$$

After extrapolation, $f(\bar{\alpha})$ is re-evaluated and the phase field equation is solved again before the next cycle block is entered. The jump size ΔN is a user-defined parameter: smaller values improve accuracy at increased cost, whereas larger values accelerate the computation but may introduce extrapolation error during stages of rapid crack growth. In practice, the strategy often reduces computational time by approximately a factor of ΔN , but the resulting error is problem-dependent and must be checked for the chosen geometry, material model, and crack-growth regime. The quantitative sensitivity to ΔN is therefore assessed explicitly in the fatigue benchmarks rather than assumed a priori.

3. Software architecture and numerical implementation

For a comprehensive overview of phase-field models for fracture and fatigue, including their solution strategies and discretisation techniques, the reader is referred to the recent review by Li et al. [49]. The present section focuses on the software architecture and the implementation choices specific to the proposed Julia/Gridap library, and complements rather than reproduces the broad numerical-implementation discussion already available in that review.

3.1. Layered library architecture

The library's design follows a single organising principle: all four physical regimes share the same variational skeleton, with differences reduced to which physical components are activated. The code is therefore structured with elastic fracture logic as the shared base, onto which plasticity and fatigue modules are layered. This allows higher-level modules to fully reuse the solver, assembly logic, and output interface of lower-level modules; adding a new physical mechanism requires only new state variables and update functions, with no changes to existing code.

The inter-module dependencies are organised as follows. `energy.jl` implements the Amor and spectral elastic strain-energy decompositions. `phasefield.jl` depends on `energy.jl` for the tensile energy and implements AT1 and AT2 phase-field subproblem assembly and solution, together with fatigue degradation and history-variable updates. `plasticity.jl` implements the quadrature-point return mapping and produces the effective stress, plastic strain tensor, and equivalent plastic strain. `solver.jl` depends on the modules above and handles elastic and elastoplastic regimes by checking whether a `PlasticStateData` object is provided. `materials.jl` manages the material-parameter structs and hardening-law dispatch. `projection.jl` performs L^2 projection of quadrature-point fields to nodes for visualisation output. `gui.jl` and `visualization.jl` provide the real-time interactive interface and offline contour-plot output, respectively. All example scripts reside in `examples/` and each script provides only the geometry, material parameters, and loading protocol at the top level; the solver and output logic are provided by the library modules.

Julia's multiple dispatch is the key technical mechanism that realises this layered architecture. The same function name automatically dispatches to different implementations based on the types of the arguments passed, with no need for explicit conditional branching or inheritance hierarchies. For elastic strain energy computation, `tensile_energy` selects the specific implementation based on a decomposition-type parameter:

```
1 tensile_energy(ee, mat, ::Val{:Amor})    # Amor
   volumetric-deviatoric split
2 tensile_energy(ee, mat, ::Val{:Spectral}) # Miehe
   spectral split
```

The `solve_step!` function distinguishes elastic and elastoplastic solution paths by checking whether a `PlasticStateData` object is provided; the calling code is identical in both cases, and the physical behaviour is determined by the type of the state object that is passed. This dispatch mechanism incurs little overhead when switching between physical regimes and avoids deep inheritance hierarchies.

Gridap's high-level weak-form API makes the mapping from formulae to code nearly verbatim. The displacement bilinear form (Eq. (10)) reads:

```
1 a_u(u, v) = integral( g_d * (C odot eps(u)) odot eps(v)
   ) dOmega
2 l_u(v)    = integral( g_d * (C odot eps_p) odot eps(v) )
   dOmega + integral( t_N dot v ) dGamma_N
```

where `g_d` corresponds to the degradation function $g(d)$, `C` to the elasticity tensor $\mathbb{C}$, `eps_p` to the quadrature-point plastic strain ϵ^p , and `t_N` to the boundary traction t . The AT2 phase field bilinear form (Eq. (14)) is correspondingly:

```
1 a_d(d, q) = integral( (f_val*Gc/10 + 2*H_field)*d*q
   + f_val*Gc/10*(grad(d) dot grad(q)) ) dOmega
2 l_d(q)    = integral( 2*H_field*q ) dOmega
```

where `H_field` corresponds to the crack-driving history variable $\mathcal{H}$ and `f_val` to the fatigue degradation value $f(\bar{a})$. The near-identity between mathematical notation and code identifiers means that researchers modifying a constitutive assumption can directly locate the relevant code without understanding the underlying matrix assembly logic. All external dependencies are recorded in `Project.toml` and reproducible with a single `Pkg.instantiate()` call; the principal packages used by the library and their roles are listed in [Appendix A.2](#) (Table A.1).

The same dispatch mechanism also provides a route to other phase-field formulations beyond AT1 and AT2. In particular, cohesive phase-field models such as the unified PF-CZM framework [50] can be added through a new dispatch tag in `phasefield.jl`, with the associated geometric crack function $w(d)$, energetic degradation function $g(d)$, and cohesive law parameters supplied through `materials.jl`; further details are given in [Appendix A.6](#).

3.2. Staggered solver

The coupled system (Eqs. (10) and (14)/(16)) is solved by a staggered alternate-minimisation scheme, in which one field is held fixed while the other is updated within each load step until convergence is reached. The principal reason for choosing this format over a fully coupled one is that the latter requires assembling a joint tangent stiffness matrix for the displacement and phase fields; once plastic coupling is activated, a consistent linearisation of this matrix becomes cumbersome and strongly inhibits constitutive modification. By contrast, the staggered format decouples the subproblems so that each remains linear when the others are prescribed, thereby keeping both assembly and solution logic transparent and readily extensible. For the quasi-static problems considered here, this strategy provides robust convergence under appropriate step-size control.

[Fig. 3](#) provides a schematic illustration of this alternating update sequence. Its role here is purely explanatory: the figure is included only to visualise the order in which the displacement field, phase field, and constitutive-state update are advanced within each increment before the cycle is repeated to convergence.

The nonlinearity introduced by plasticity is handled by an outer Picard fixed-point iteration embedded within this staggered loop. Within each load substep, given the current damage field d and the plastic strain $\epsilon_{(k)}^p$ from the previous Picard iterate, the displacement subproblem is solved for the new displacement field $u_{(k+1)}$; the return-mapping algorithm (Section 3.3) then updates the plastic state at each quadrature point; the two steps are repeated until convergence. The Picard iteration avoids the need for a consistent tangent stiffness matrix, at the cost of linear rather than quadratic convergence, but for the hardening materials considered here convergence is typically achieved within 3–5 iterations.

Crack irreversibility ($d_{n+1} \geq d_n$) is enforced after each phase-field solve by a pointwise clamp $d_{n+1} \leftarrow \max(d_{n+1}, d_n)$; the literal Julia routine is given in [Appendix A.7](#). This clamp is preferred over penalty or augmented-Lagrange formulations because the phase-field equation already guarantees monotonic damage growth through the monotonicity of $\mathcal{H}$, and the clamp therefore acts only as a safe projection step that does not affect the residual and introduces no additional parameters.

The crack-driving history variable $\mathcal{H}$ (Eq. (27)) is updated at each quadrature point by a pointwise max operation, $\mathcal{H} \leftarrow \max(\mathcal{H}, \psi^{e+} + \beta\psi^p)$, whose implementation is also given in [Appendix A.7](#). The max operation guarantees monotonicity directly, with no additional state-management logic required. The complete staggered solution procedure including the four-regime branching and cycle-jump acceleration is given in Algorithm 1. Both linear subsystems are solved by the direct sparse solver MUMPS; finite element spaces, sparse matrix patterns, quadrature-point caches, and boundary-condition objects are created once during initialisation and reused across all load steps, avoiding repeated allocation overhead. All benchmark computations in this work

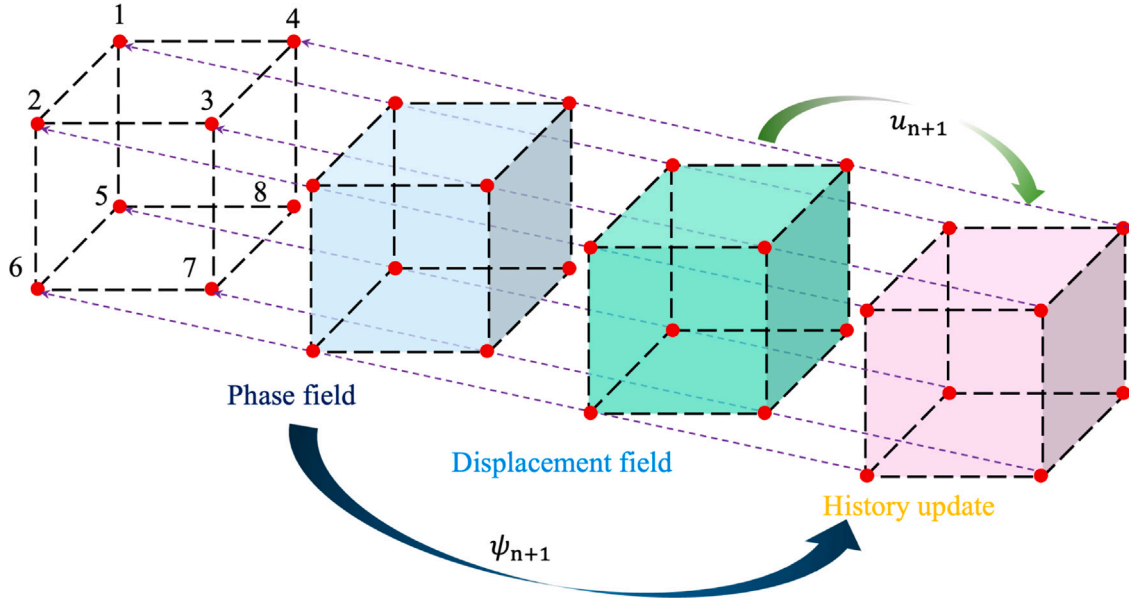


Fig. 3. Schematic illustration of the staggered solution procedure. Within each load increment, the displacement field, phase field, and constitutive-state update are solved in sequence and iterated until convergence. The figure is used here only to visualise the alternating update structure of the solver.

Algorithm 1 Staggered phase field solver with four-regime branching and cycle-jump acceleration.

Require: Mesh, material parameters, G_c , l_0 , α_T , β , ΔN , N_{total}
Ensure: Phase field d , displacement u , crack/load history

- 1: Initialise: $d = 0$, $\tilde{\alpha} = 0$, $\mathcal{H} = 0$, $\epsilon^p = \mathbf{0}$, $\alpha_p = 0$, $N = 0$
- 2: **while** $N < N_{\text{total}}$ **do**
- 3: $N \leftarrow N + \Delta N$ ▷ Start new cycle block
- 4: **for** each substep t_i in one representative cycle **do**
- 5: Solve displacement (10) via Picard iteration $\rightarrow u$
- 6: **if** plastic regime active **then**
- 7: Return mapping at each QP $\rightarrow \tilde{\sigma}$, ϵ^p , α_p ; compute ψ^p
- 8: **end if**
- 9: Compute ψ^{e+} (Amor or spectral)
- 10: $\mathcal{H} \leftarrow \max\{\mathcal{H}, \psi^{e+} + \beta \psi^p\}$
- 11: **if** fatigue regime active **then**
- 12: $\tilde{\alpha} \leftarrow g(d)[\psi^{e+} + \beta \psi^p]$; $\Delta \tilde{\alpha}_i \leftarrow \max(0, \tilde{\alpha}_i - \tilde{\alpha}_{i-1})$
- 13: **end if**
- 14: **end for**
- 15: **if** fatigue regime active **then**
- 16: $\tilde{\alpha} \leftarrow \tilde{\alpha} + \Delta N \cdot \sum_i \Delta \tilde{\alpha}_i$ ▷ Cycle-jump (Eq. (31))
- 17: Evaluate $f(\tilde{\alpha})$ ▷ Eq. (30)
- 18: **end if**
- 19: **repeat**
- 20: Solve phase field (14)/(16) $\rightarrow d^{(k+1)}$
- 21: $d^{(k+1)} \leftarrow \max(d^{(k+1)}, d^{(k)})$ ▷ Irreversibility
- 22: **until** $\|d^{(k+1)} - d^{(k)}\| / \|d^{(k+1)}\| < 10^{-3}$
- 23: Record output: (N, a) or (u, F)
- 24: **end while**

are performed on a workstation with an Intel Core i9-14900K (24 cores, 5.6 GHz boost) and 64 GB DDR5 memory.

Algorithm 1 provides the complete solution procedure for all four physical regimes with cycle-jump acceleration.

3.3. Radial return mapping

The return-mapping algorithm is executed independently at each quadrature point in the full 3D effective stress space. Under plane-strain conditions, although the macroscopic problem is solved in 2D,

the complete three-dimensional stress and strain state at each quadrature point must still be tracked explicitly; this point is discussed in Section 2.3. Restricting the algorithm to a 2D plane-stress space would incorrectly set the out-of-plane normal stress σ_{33} to zero and introduce systematic errors in the von Mises equivalent stress and in the plastic stored energy density ψ^p . The present library therefore executes the return mapping in full 3D, which is consistent with the plane-strain plasticity assumption.

Given the converged plastic state from the previous step (ϵ_n^p , $\alpha_{p,n}$) and the current total strain ϵ_{n+1} , the algorithm first computes the elastic trial stress:

$$\epsilon^{e,\text{tr}} = \epsilon_{n+1} - \epsilon_n^p, \quad \tilde{\sigma}^{\text{tr}} = \mathbb{C} : \epsilon^{e,\text{tr}} \quad (32)$$

If the trial yield function $F^{\text{tr}} = \tilde{\sigma}_{\text{eq}}^{\text{tr}} - \sqrt{2/3} \kappa(\alpha_{p,n}) \leq 0$, the elastic step is accepted; otherwise the scalar equation $R(\Delta\gamma) = 0$ is solved for the plastic multiplier increment:

$$R(\Delta\gamma) = \tilde{\sigma}_{\text{eq}}^{\text{tr}} - 3G\Delta\gamma - \sqrt{\frac{2}{3}} \kappa \left(\alpha_{p,n} + \sqrt{\frac{2}{3}} \Delta\gamma \right) = 0 \quad (33)$$

For linear hardening, the closed-form solution $\Delta\gamma = F^{\text{tr}} / (3G + \frac{2}{3}H)$ exists. For power-law hardening, a local Newton–Raphson iteration is used, with the linear-hardening closed-form solution as the initial guess (using the tangent hardening modulus at the current equivalent plastic strain to approximate H); this strategy reduces the typical iteration count from 8–10 to 3–5, with convergence criterion $|R(\Delta\gamma)| < 10^{-12}$. The updated effective stress is

$$\tilde{\sigma}_{n+1} = \tilde{\sigma}^{\text{tr}} - 2G\Delta\gamma \mathbf{N}^{\text{tr}}, \quad \mathbf{N}^{\text{tr}} = \frac{3}{2} \frac{\text{dev}(\tilde{\sigma}^{\text{tr}})}{\tilde{\sigma}_{\text{eq}}^{\text{tr}}} \quad (34)$$

The degradation function $g(d)$ is applied only after the return mapping is complete, to obtain the nominal stress $\sigma = g(d)\tilde{\sigma}$; it does not enter any intermediate step of the return mapping, consistent with the effective-stress-space plasticity formulation of Section 2.4.

The Julia implementation, in which variable names are in direct correspondence with the mathematical notation above, is given in Appendix A.7.

3.4. Cycle-jump coupling with the staggered solver

The mathematical formulation of the cycle-jump strategy is given in Section 2.5.2; this subsection focuses on how it is coupled with

the staggered solver. In the outer loop of Algorithm 1, each iteration corresponds to a cycle block rather than a single cycle: one representative cycle of N_{sub} substeps is first simulated in full, during which displacement solving, return mapping (if plasticity is active), elastic strain energy, and plastic stored energy are all computed, and the per-cycle fatigue increment is accumulated substep-by-substep via Eq. (29). At the end of the cycle this increment is extrapolated over ΔN cycles using Eq. (31); the corresponding update routine is given in Appendix A.7. After extrapolation, the phase-field equation is re-solved to update the damage field before entering the next cycle block. The extrapolation step updates only $\bar{a}$ and $f(\bar{a})$; the displacement field and plastic state are updated in the next representative-cycle solve, where the effect of damage-field changes on the structural response is captured. This coupling preserves consistency between fatigue degradation and structural response while reducing the computational cost by a factor of approximately ΔN .

The choice of ΔN directly affects prediction accuracy and computational efficiency. In the early stage when fatigue damage accumulates slowly, larger values (e.g., $\Delta N = 100\text{--}500$) can be used safely; in the rapid crack-propagation stage, the per-cycle increment of $\bar{a}$ grows substantially, and ΔN should be reduced to prevent extrapolation errors from accumulating. All benchmark cases in this work use a fixed ΔN ; its quantitative effect on $a\text{--}N$ curve accuracy is reported in the respective benchmark sections. Adaptive ΔN strategies are left for future work.

4. Numerical benchmarks

This section validates the library across four physically distinct fracture and fatigue regimes. Eight benchmark cases in total are presented, with two per physical regime. Cases that scan multiple parameter values (e.g., three hardening moduli in Section 4.2.1 or three notch-offset configurations in Section 4.3.2) are counted as single benchmark problems, since they share identical geometry, boundary conditions, and solver settings and differ only in one parameter. Two representative benchmark cases are presented per subsection, arranged in order of increasing physical complexity: elastic fracture, ductile fracture, elastic fatigue, and elastoplastic fatigue. All simulations are plane-strain unless stated otherwise; material and phase field parameters are reported in the same units as the respective reference sources. The AT2 model (11) with the Amor volumetric–deviatoric energy split is used in Sections 4.1 and 4.2; the AT1 model (12) with the length-scale-insensitive formulation of Mandal et al. [51] is adopted for the fatigue benchmarks in Sections 4.3 and 4.4.

4.1. Elastic fracture

The two canonical Miehe–Hofacker–Welschinger benchmarks [6] are reproduced on the same $1 \times 1 \text{ mm}^2$ single-edge-notched geometry to verify the implementation for pure Mode I tension and mixed-mode shear loading respectively. Both cases use $\eta_k = 10^{-10}$ and $\varepsilon^p \equiv 0$ ($\beta = 0$).

4.1.1. Single-edge-notched tension (SENT)

The SENT specimen is a $1 \times 1 \text{ mm}^2$ square plate with a horizontal pre-crack of length $a_0 = 0.5 \text{ mm}$ extending from the left edge at mid-height [6]. Material and phase field parameters are listed in Table 4. The bottom edge is fully clamped ($u_x = u_y = 0$) and the top edge is loaded in pure vertical tension; the displacement increment is $\Delta u = 8 \times 10^{-6} \text{ mm}$ throughout, resolving the snap-back response. The mesh is refined to element size $h = 0.005 \text{ mm}$ along the crack band.

Fig. 4 shows the damage contours and the predicted $F\text{--}u$ curve. The crack nucleates at the pre-crack tip and propagates horizontally across the full specimen width in pure Mode I; the damage band remains sharply localised within the refined mesh region, with no spurious damage elsewhere. The peak load is approximately 0.69 kN and is reached at $u \approx 5.4 \times 10^{-3} \text{ mm}$, followed immediately by a rapid post-peak snap-back, both in quantitative agreement with Miehe et al. [6] and

Table 4

Material and phase field parameters for the SENT/SENS elastic fracture benchmarks [6].

Parameter	Symbol	Value	Unit
Young's modulus	E	210 000	MPa
Poisson's ratio	ν	0.300	–
First Lamé constant	λ	121 150	MPa
Shear modulus	G	80 770	MPa
Critical energy release rate	G_c	2.7	N/mm
Regularisation length	l_0	0.015	mm
Stabilisation	η_k	10^{-10}	–

the independent phasefieldx benchmark [35]. The Amor energy decomposition correctly excludes the compressive energy component from the crack-driving force, so the compressive stiffness of the specimen is preserved up to fracture.

4.1.2. Single-edge-notched shear (SENS)

The SENS benchmark [6] uses identical geometry and material parameters to the SENT case (Table 4) but applies horizontal shear displacement to the top edge ($u_x = u$, $u_y = 0$) while the left edge is additionally constrained vertically ($u_y = 0$) to prevent rigid rotation. This loading activates mixed Mode I/II crack growth that curves from the pre-crack tip towards the lower-right corner, making the benchmark a stringent test for the energy decomposition.

Fig. 5 presents the damage contours and the $F\text{--}u$ curve. Without energy decomposition, purely compressive zones ahead of the crack tip would drive spurious damage, yielding an incorrect horizontal path. The Amor split correctly partitions elastic energy into dilatational and deviatoric components: only the former contributes to the crack-driving force, so damage remains confined to the tensile lobe of the stress field and the crack follows the experimentally observed downward-curving trajectory. The spectral split produces a nearly identical path, confirming that both decompositions are suitable for this mixed-mode geometry. The predicted peak shear load $F_{\text{peak}} \approx 0.45 \text{ kN}$ and the overall shape of the $F\text{--}u$ curve agree with the reference. The ability to reproduce complex mixed-mode paths without any predefined trajectory is a key strength of the variational phase field approach.

4.2. Ductile fracture

Two monotonic ductile fracture benchmarks validate the coupled $J_2\text{--}AT2$ formulation introduced in Section 2.4. Plane-strain conditions are enforced throughout; the three-dimensional radial return mapping of Section 3.3 is used to obtain the correct volumetric plastic response under plane-strain kinematics.

4.2.1. Double-notched monotonic ductile fracture

The benchmark geometry is a $10 \times 10 \text{ mm}^2$ square plate with two semicircular notches positioned at diagonally opposite corners: the bottom-left notch has radius $r_1 = 2.5 \text{ mm}$ and the top-right notch $r_2 = 2.0 \text{ mm}$, following the specification of Ambati et al. [5] Section 5.5. Material and phase field parameters are listed in Table 5. The mesh is refined to $h_{\text{fine}} = 0.07 \text{ mm}$ along the diagonal connecting the two notch roots.

Two loading scenarios are investigated by simultaneously displacing the top and left edges while fixing the bottom and right edges. In the $u_L/u_T = 1.0$ case, equal vertical and horizontal displacements are applied, corresponding to biaxial tension. In the $u_L/u_T = 0.7$ case, the horizontal component is reduced, creating a preferential tensile direction that biases the crack trajectory towards a more vertical path. The total applied displacement is $u_T = 0.40 \text{ mm}$.

Fig. 6 presents the benchmark geometry, the two force–displacement curves, and the full history of damage and equivalent plastic strain for both loading cases. In both cases, a broad plastic zone forms and expands across the ligament well before any measurable damage

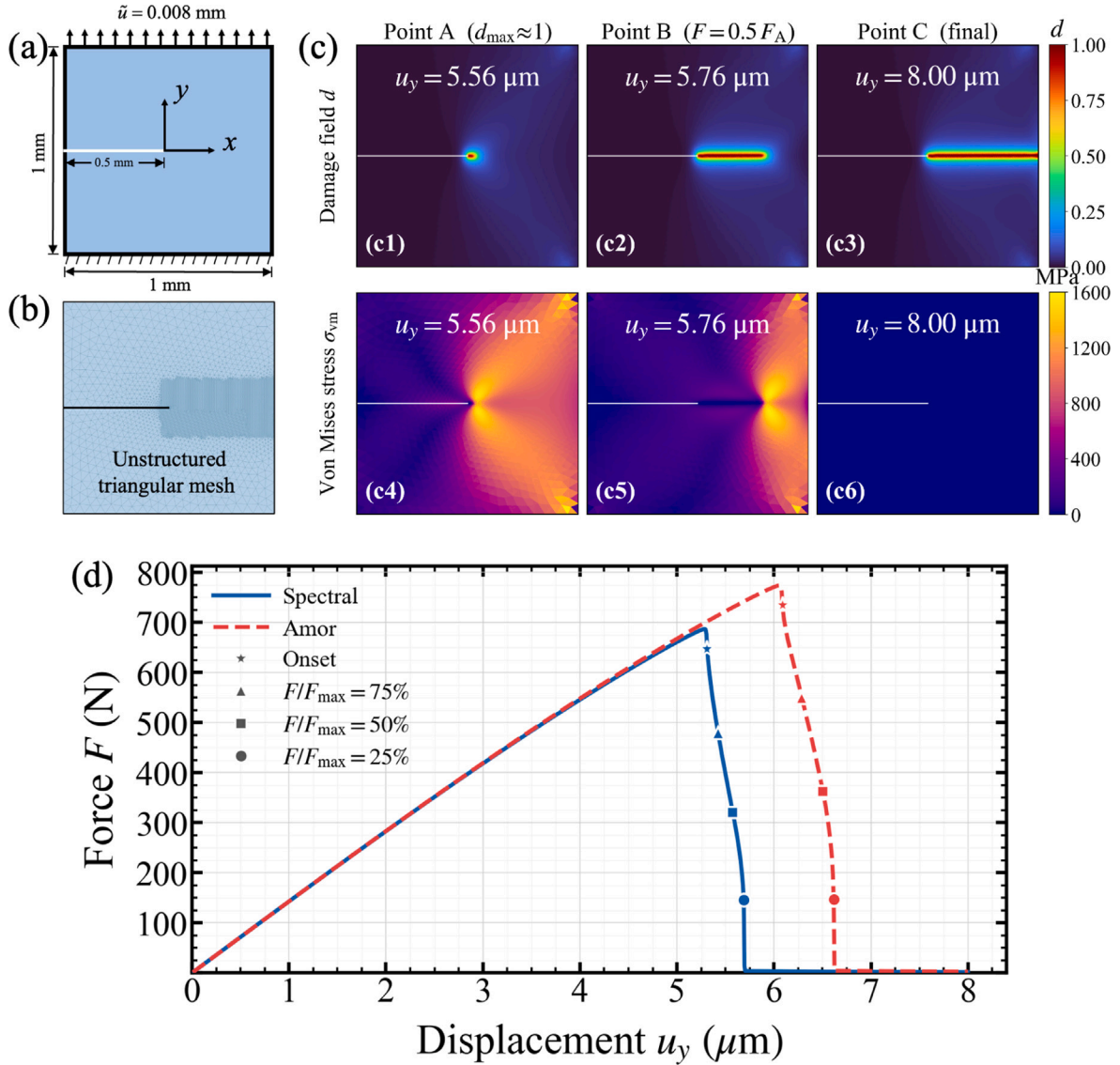


Fig. 4. SENT elastic fracture (Miehe et al. [6] Section 5.1): (a) specimen geometry and loading/boundary-condition schematic; (b) finite-element mesh; (c1)–(c3) phase field damage d and (c4)–(c6) von Mises equivalent stress σ_{eq} at instants A (pre-peak), B (peak), and C (post-peak); (d) force–displacement curve compared with the reference solution [6] and the phasefieldx benchmark [35]. The crack nucleates at the pre-crack tip and propagates horizontally in pure Mode I without any prescribed trajectory.

Table 5

Material and phase field parameters for the double-notched monotonic ductile fracture benchmark (Ambati et al. [5] Material II).

Parameter	Symbol	Value	Unit
Young's modulus	E	180 140	MPa
Poisson's ratio	ν	0.280	–
Bulk modulus	K	136 500	MPa
Shear modulus	G	70 300	MPa
Initial yield stress	σ_{y0}	443	MPa
Linear hardening modulus	H	300	MPa
Critical energy release rate	G_c	20.9	N/mm
Regularisation length	l_0	0.03	mm
Taylor–Quinney coefficient	β	1.0	–

initiates: this is the hallmark of ductile fracture, in which extensive plastic deformation precedes fracture initiation. At peak load the plastic

strain ϵ_{eq}^p reaches values of order 10%–20% throughout the ligament, whereas the damage field remains localised near the notch roots. Post-peak, the damage band propagates rapidly through the pre-plastified matrix and the load drops sharply.

The $u_L/u_T = 0.7$ loading condition modifies both the crack trajectory and the global response. Reducing the horizontal displacement component rotates the principal tensile direction, so the dominant crack path curves more steeply towards the vertical and the two notch-root damage zones merge at a slightly different location compared to the $u_L/u_T = 1.0$ case. The peak force is lower and the fracture displacement is slightly reduced, because the larger effective Mode I component aligned with the crack trajectory accelerates damage evolution through the crack-driving history variable $\mathcal{H}$. These trends are in qualitative and quantitative agreement with the reference results of Ambati et al. [5], confirming that the staggered Picard iteration correctly captures the coupling between J_2 plasticity and phase field damage in the ductile regime.

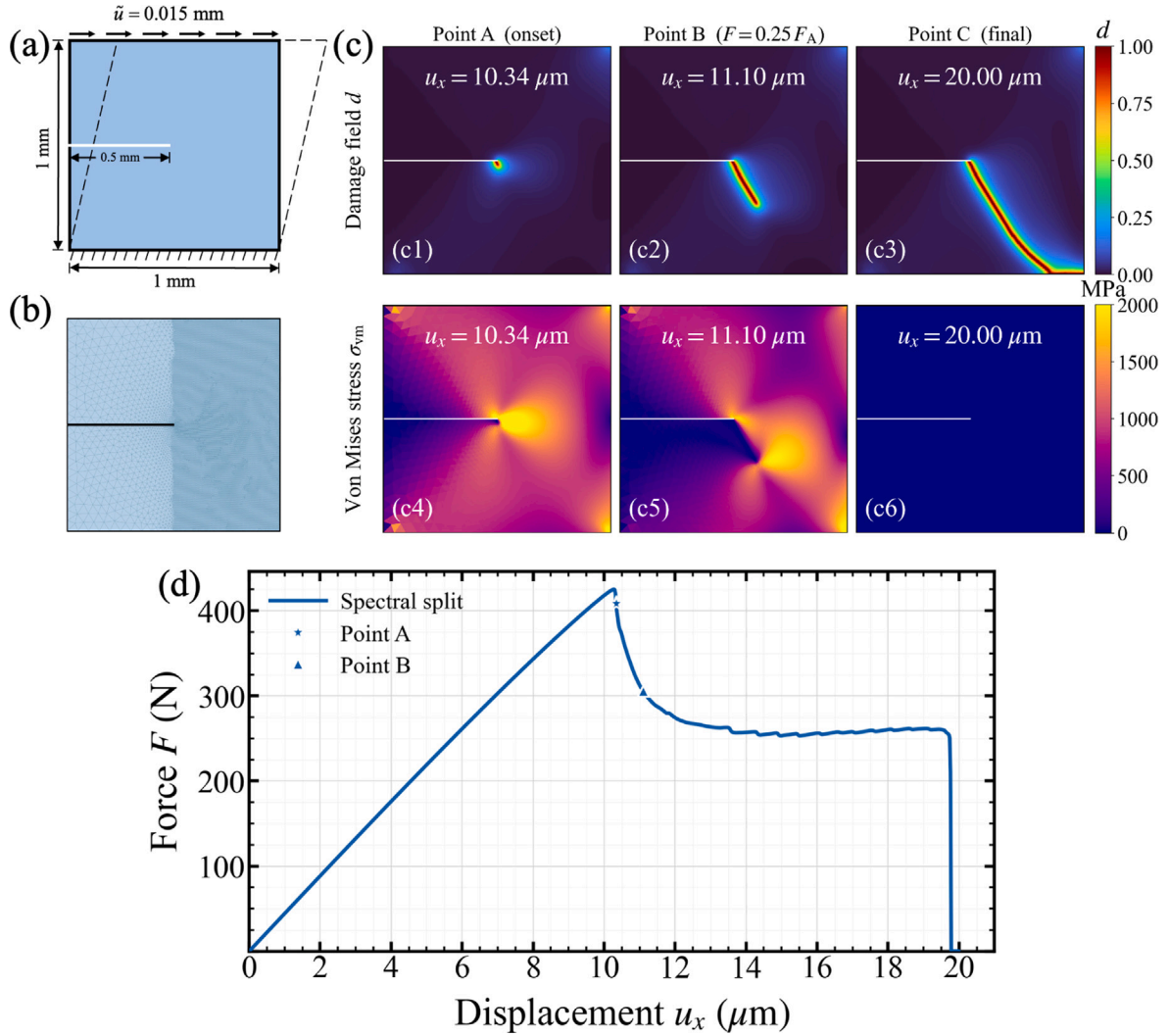


Fig. 5. SENS elastic shear fracture (Miehe et al. [6] Section 5.2): (a) specimen geometry and loading/boundary-condition schematic; (b) finite-element mesh; (c1)–(c3) phase field damage d and (c4)–(c6) von Mises equivalent stress σ_{eq} at three loading instants; (d) force–displacement curve compared with the reference solution. The crack follows a curved mixed-mode path without any prescribed trajectory; both Amor and spectral energy splits reproduce the correct deflection direction, with only minor differences in the crack angle.

4.2.2. Flat-grooved specimen: effect of hardening modulus

The second ductile fracture benchmark explores the sensitivity of the coupled plastic–damage response to the isotropic hardening modulus. The specimen is an 18×50 mm² rectangular plate with two semicircular notches of radius $r = 2.5$ mm on opposite sides at staggered heights: the left notch is centred at $y = 30$ mm and the right notch at $y = 20$ mm, introducing a vertical offset of 10 mm that drives an asymmetric inter-notch crack path. Material and phase field parameters are listed in Table 6. Three hardening ratios are compared: $H = 0.1E$, $H = 0.2E$, and $H = 0.5E$, corresponding to $H = 7148$, 14 296, and 35 740 MPa respectively. The specimen is loaded monotonically in vertical tension to a total displacement of 0.30 mm.

Fig. 7 shows the specimen geometry and the finite-element mesh used for the three hardening ratios. Fig. 8 plots the force–displacement response for the three hardening levels, and Fig. 9 shows the corresponding damage and equivalent plastic strain fields. At low hardening ($H = 0.1E$) the material strain-hardens slowly, the plastic zone remains relatively narrow, and the crack propagates along a near-direct path from the left notch root to the right notch root across the 10 mm offset. The fracture surface is smooth and the load–displacement peak is sharp and well-defined. At intermediate hardening ($H = 0.2E$) the plastic zone is wider and redistributes the crack-driving energy more

Table 6

Material and phase field parameters for the flat-grooved specimen ductile fracture benchmark. Three hardening moduli are scanned ($H/E = 0.1, 0.2, 0.5$); all other parameters are fixed.

Parameter	Symbol	Value	Unit
Young's modulus	E	71 480	MPa
Poisson's ratio	ν	0.3	–
Initial yield stress	σ_{y0}	345	MPa
Hardening modulus (scan)	H	0.1E/0.2E/0.5E	MPa
Critical energy release rate	G_c	9.31	N/mm
Regularisation length	l_0	0.2	mm
Taylor–Quinney coefficient	β	1.0	–

evenly; the crack deflects slightly before merging with the damage zone emanating from the right notch. At high hardening ($H = 0.5E$) the plastic zone is sufficiently broad to suppress rapid crack propagation: the two damage bands grow slowly and spread laterally before coalescing, and the post-peak softening is noticeably more gradual than at lower H . Throughout the three cases, the peak load increases monotonically with H/E , and so does the fracture displacement. This behaviour is a direct consequence of the β -weighted crack-driving force: larger H stores more energy in the plastic field before the phase field threshold

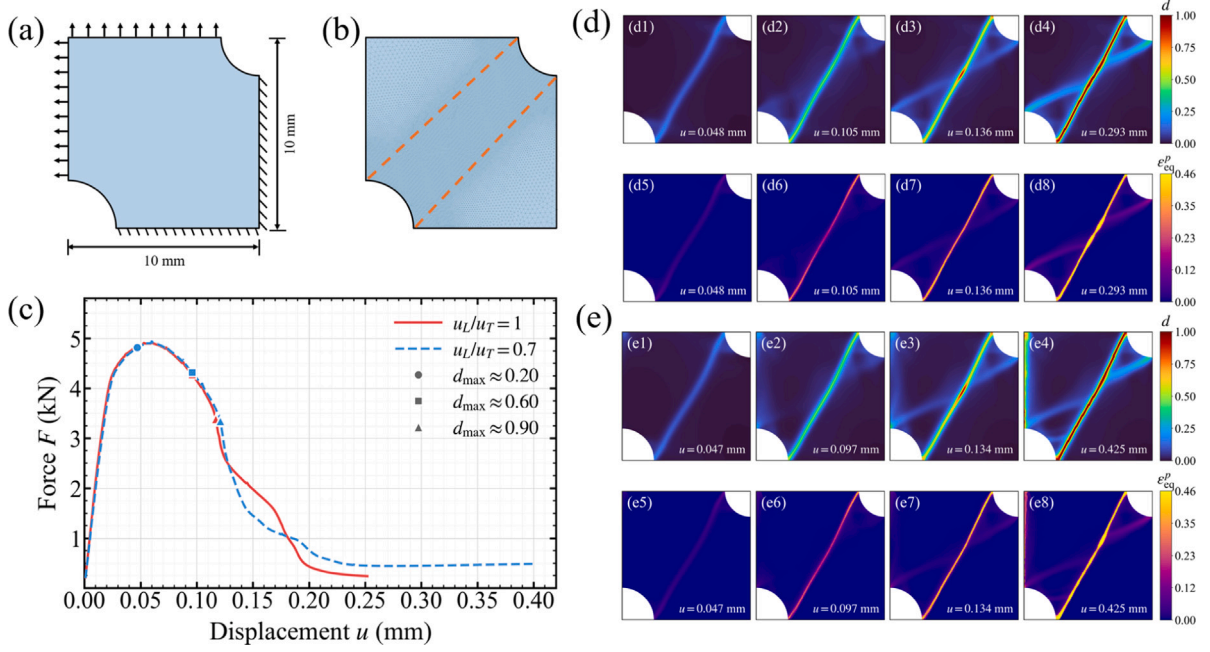


Fig. 6. Double-notched monotonic ductile fracture (Ambati et al. [5], Section 5.5). (a) Geometry and boundary conditions for the $u_L/u_T = 1.0$ loading case. (b) Geometry, mesh, and crack-path sketch for the $u_L/u_T = 0.7$ loading case. (c) Force–displacement curves for the two loading cases, with markers indicating representative states with $d_{\max} \approx 0.20$, 0.60 , and 0.90 . (d) Phase field damage d (top row) and equivalent plastic strain ϵ_{eq}^p (bottom row) at four loading instants for $u_L/u_T = 1.0$. (e) Corresponding damage and plastic-strain fields for $u_L/u_T = 0.7$. Reducing the horizontal load component steepens the dominant crack path, lowers the peak force, and causes slightly earlier fracture initiation.

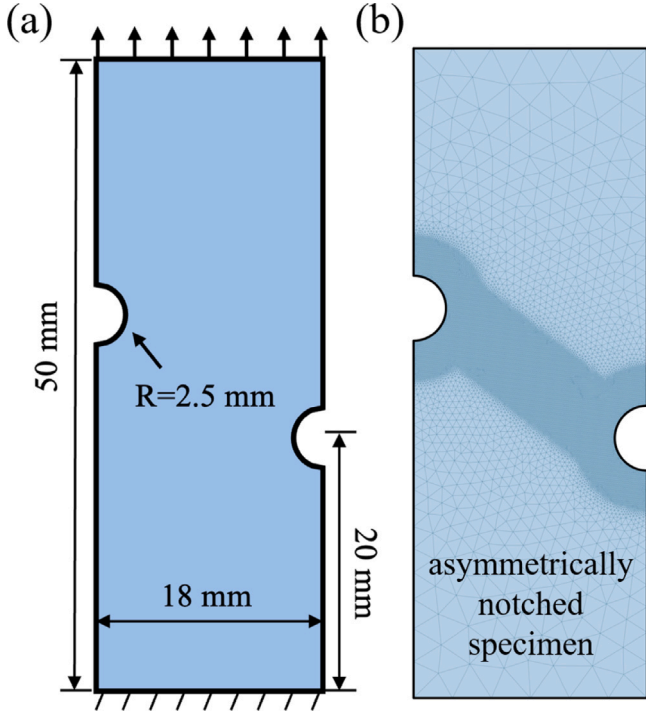


Fig. 7. Flat-grooved specimen ductile fracture: specimen geometry and finite-element mesh. (a) Geometry of the asymmetrically notched specimen, with two semicircular notches of radius $R = 2.5$ mm on opposite sides at staggered heights, introducing a 10 mm vertical offset between the notches. The bottom edge is fully clamped and the top edge is loaded in vertical tension. (b) Finite element mesh with local refinement along the expected inter-notch ligament.

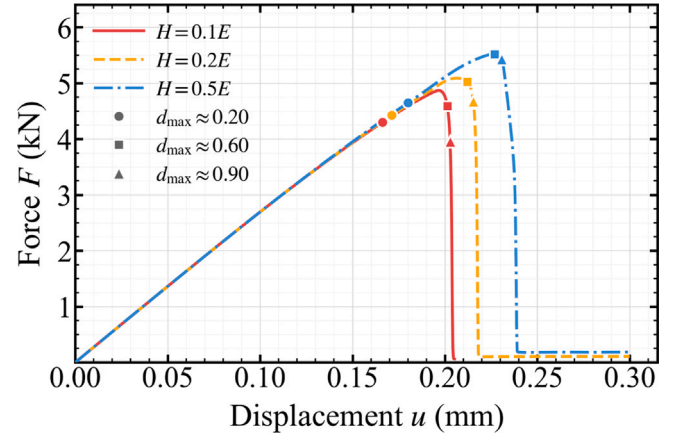


Fig. 8. Force–displacement curves for the flat-grooved ductile fracture specimen for three hardening ratios, $H/E = 0.1$, 0.2 , and 0.5 . The circular, square, and triangular markers indicate representative states with $d_{\max} \approx 0.20$, 0.60 , and 0.90 , respectively, used to select the field snapshots. Increasing the hardening modulus raises both the peak force and the fracture displacement, while delaying unstable crack initiation.

is exceeded, effectively increasing the apparent fracture resistance. The results are consistent with the general trend reported in the ductile phase field fracture literature [22,23].

The accompanying changes in peak force, fracture displacement, plastic-zone width, and inter-notch crack-path curvature can be understood as different manifestations of the same hardening-induced increase in the β -weighted crack-driving force. Higher hardening allows the post-yield stress to keep growing with plastic strain, so both the elastic positive energy and the plastic stored energy continue to accumulate at higher load levels and at larger applied displacement before the phase-field threshold is exceeded; this directly raises the peak force and the fracture displacement. At the same time, suppressed

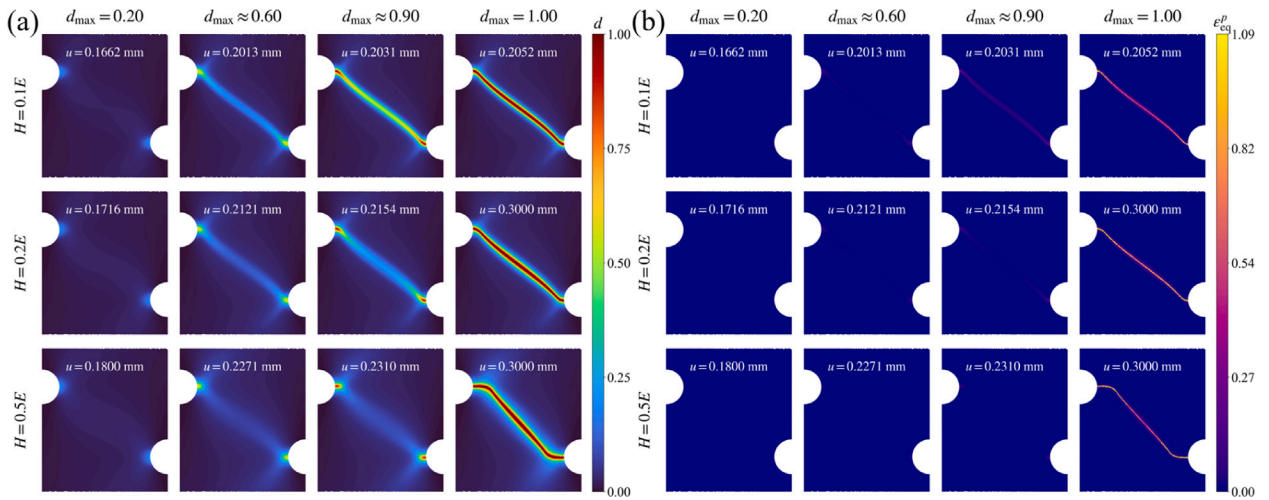


Fig. 9. Flat-grooved specimen ductile fracture: damage and equivalent plastic strain fields for three hardening ratios, $H/E = 0.1, 0.2$, and 1.0 . (a) Phase field damage d at four representative damage levels ($d_{\max} \approx 0.20, 0.60, 0.90$, and 1.00) for each hardening ratio. (b) Corresponding equivalent plastic strain ϵ_{eq}^p fields at the same instants. Stronger hardening broadens the plastic zone, delays crack localisation, and produces a more curved inter-notch crack path.

Table 7

Shared material and phase field parameters for the elastic fatigue benchmarks (AT1 length-scale-insensitive model [51]). The regularisation length l_s differs between the two cases; all other entries are common.

Parameter	Symbol	Value	Unit
Young's modulus	E	20.8	GPa
Poisson's ratio	ν	0.3	–
Critical energy release rate	G_c	5×10^{-4}	N/mm
Critical stress	σ_c	0.2	GPa
Fatigue threshold factor	k_f	0.5	–
Fatigue threshold	$\sigma_T = \sigma_c^2 / (2Ek_f)$	$\approx 1.92 \times 10^{-3}$	GPa
Regularisation length (TPB)	l_s	0.06	mm
Regularisation length (offset)	l_s	0.10	mm

stress saturation lets the post-yield stress field spread over a larger volume, so a wider region around the notch tips reaches yield and the equivalent plastic strain field broadens accordingly, as seen in Fig. 9(b). The resulting driving-force distribution is more diffuse, with a flatter gradient between the two notches; the damage band is therefore no longer attracted strongly to the shortest direct ligament and curves laterally as H/E increases, which is consistent with the progression from a near-direct path at $H = 0.1E$ to a more curved path at $H = 0.5E$ in Fig. 9(a).

4.3. Elastic fatigue

Two elastic fatigue benchmarks are presented using the AT1 length-scale-insensitive formulation [51] with the cycle-jump acceleration scheme. In both cases the specimen is loaded by displacement-controlled cyclic loading at the top edge; the loading waveform and the physical interpretation of the fatigue driving force are illustrated in Fig. 10. The phase field subproblem is solved at every represented cycle, while the cumulative fatigue variable $\bar{a}$ is extrapolated over ΔN cycles in each jump to reduce computational cost [13,15]. Both benchmarks share the same elastic material constants and phase field toughness; they differ only in the regularisation length l_s , which is 0.06 mm for the three-point bending case and 0.10 mm for the offset double-notch case. Common parameters are listed in Table 7.

4.3.1. Three-point bending elastic fatigue (AT1 model)

A three-point bending beam of $L = 8.0$ mm and height $H = 2.0$ mm with a central notch of width 0.2 mm and depth 0.4 mm is loaded cyclically at its midspan. The notch tip is offset slightly from the exact

centre ($x/L \in [0.475, 0.525]$) to regularise the symmetry. Material parameters are given in Table 7 with $l_s = 0.06$ mm. The AT1 length-scale-insensitive correction factor $m = 3EG_c / (4l_s\sigma_c^2)$ is determined from G_c , l_s , and σ_c according to [51]. Loading: $u_{\max} = 0.0125$ mm, $R = 0$, $N_{\text{total}} = 400$ cycles, $\Delta N = 1$.

Fig. 11 presents the damage and fatigue-degradation contours at four snapshots over the 400-cycle loading history, cropped to the central $2H \times H$ region around the notch tip. In the early cycles, $f(\bar{a})$ decreases progressively ahead of the notch while the damage field remains below the nucleation threshold. This precursor phase corresponds to fatigue softening that accumulates without visible cracking. This two-stage behaviour is intrinsic to the AT1 model: unlike AT2, the AT1 formulation has a finite elastic threshold, so damage cannot initiate until the degraded effective toughness $f(\bar{a})G_c$ falls to the point where the crack-driving energy $\mathcal{H}$ reaches the nucleation criterion. Once this threshold is exceeded, a sharp damage band nucleates at the notch root and propagates upward in pure Mode I over a small number of additional cycles. The spatial distribution of $f(\bar{a})$ closely mirrors the concentration of elastic strain energy around the notch, confirming that the length-scale-insensitive AT1 formulation correctly redistributes the fatigue driving force without introducing spurious size dependence. The predicted crack-growth trajectory and final crack length at $N = 400$ cycles are consistent with the reference benchmark of Mandal et al. [51].

4.3.2. Offset double-notch elastic fatigue: crack-path interaction and coalescence

The second elastic fatigue benchmark investigates crack interaction and coalescence in a specimen with two edge pre-cracks at deliberately staggered heights, representing the scenario of multiple co-planar or offset defects commonly encountered in welded structures and turbine components. The geometry is an 18×50 mm² plate with left and right pre-cracks of half-length $a = 2.5$ mm. Three notch configurations are compared: *aligned* (both cracks at $y = 25$ mm, symmetric), *offset-mid* (left at $y = 27$ mm, right at $y = 23$ mm, offset $\Delta y = 4$ mm), and *offset-large* (left at $y = 29$ mm, right at $y = 21$ mm, offset $\Delta y = 8$ mm). All three cases share the material parameters of Table 7 with $l_s = 0.10$ mm (all other entries unchanged). Loading: $u_{\max} = 0.05$ mm, $R = 0.1$, $N_{\text{total}} = 400$ cycles, $\Delta N = 1$. VTU snapshots are saved every 20 cycles.

Fig. 12 presents the damage and fatigue-degradation contours and the ligament-gap growth curves. In the *aligned* case, the two crack tips advance symmetrically and coalesce along a straight horizontal ligament well within the 400-cycle window; the ligament-gap curve drops

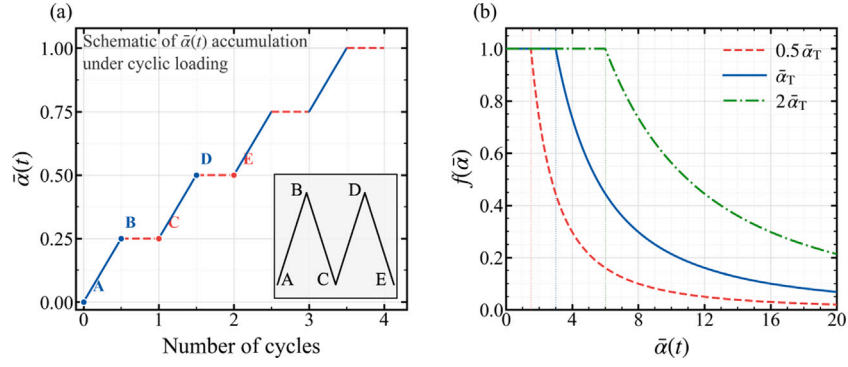


Fig. 10. Schematic of displacement-controlled cyclic loading used in the fatigue benchmarks. The applied displacement oscillates between $u_{\min} = Ru_{\max}$ and $u_{\max}$; the cumulative fatigue variable $\bar{\alpha}$ accumulates irreversibly during the loading half-cycle and drives progressive degradation of the effective fracture toughness $f(\bar{\alpha})G_c$. The load ratio R controls the mean stress level and the residual crack-closure effect.

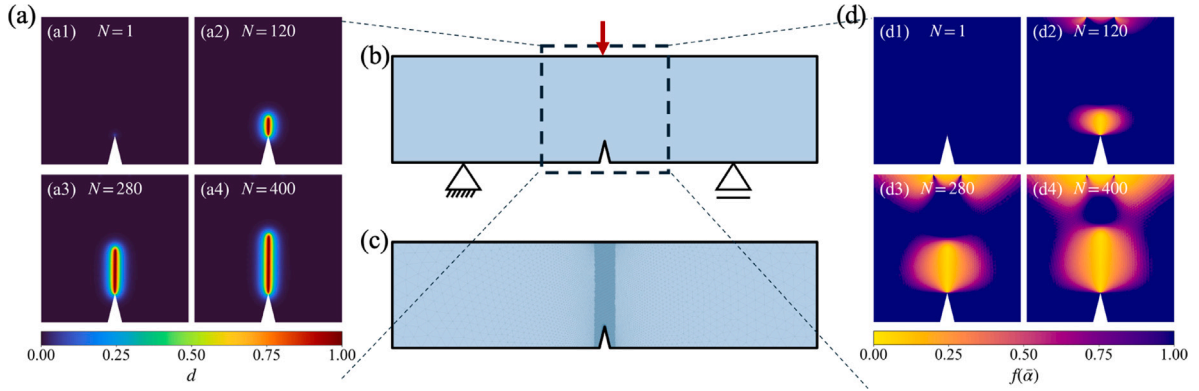


Fig. 11. Three-point bending elastic fatigue (AT1 length-scale-insensitive model [51]): (a1)–(a4) phase field damage d at four cycle counts; (d1)–(d4) fatigue degradation function $f(\bar{\alpha})$ at the same instants; view cropped to the $2H \times H$ central region. The fatigue degradation zone (rows d) precedes the damage band (rows a): toughness degrades progressively in the high-stress region at the notch tip before the damage threshold is exceeded.

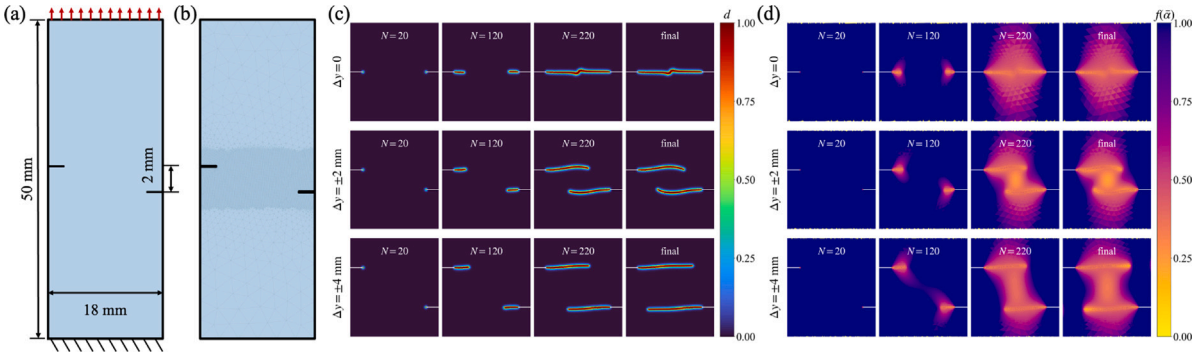


Fig. 12. Offset double-notch elastic fatigue: 3×4 contour panel showing phase field damage d (left sub-block) and fatigue degradation $f(\bar{\alpha})$ (right sub-block) for the three offset cases (rows: aligned, offset-mid, offset-large) at cycles 20, 120, 220, and final. Right panel: ligament-gap crack-growth curves for all three cases. Increasing the notch offset delays coalescence, curves the inter-crack path, and converts a direct symmetric Mode I trajectory (aligned) to an oblique mixed-mode coalescence route (offset-large).

steeply once the two damage zones merge, and the fatigue-degradation field $f(\bar{\alpha})$ exhibits a concentrated symmetric pattern centred on the mid-ligament. For the *offset-mid* case, the two crack tips are attracted by the stress concentration of the opposing tip and each curves towards it, producing the S-shaped crack trajectory characteristic of offset cracks in fracture mechanics. Coalescence is delayed relative to the aligned case because the oblique crack path is longer and the Mode I stress intensity factor is lower along the inclined segment; the ligament-gap curve shows a two-stage decline corresponding to the initial oblique approach and the final rapid link-up. In the *offset-large* case, the vertical separation is sufficient to weaken the tip-to-tip

stress interaction, and each crack advances primarily in Mode I outward from its own notch before the two damage zones eventually merge at a larger applied displacement. The corresponding ligament-gap curve is noticeably flatter in the early cycles before a sharp coalescence event. The progressive transition from symmetric direct coalescence (aligned) through S-shaped oblique coalescence (offset-mid) to near-independent propagation followed by late coalescence (offset-large) is physically consistent with classical fracture mechanics predictions and is reproduced seamlessly by the phase field model without any crack-path prescriptions or contact constraints.

Table 8

Material and phase field parameters for the CT specimen (GH4169, 550 °C) [28]. Voce isotropic hardening: $\kappa = \sigma_{y0} + Q_{\infty}(1 - e^{-b\alpha_p})$; Armstrong–Frederick kinematic hardening with modulus C and rate γ .

Parameter	Symbol	Value	Unit
Young's modulus	E	161 000	MPa
Poisson's ratio	ν	0.3	–
Initial yield stress	σ_{y0}	900	MPa
Voce saturation stress	Q_{∞}	80	MPa
Voce rate parameter	b	5	–
AF kinematic modulus	C	12 000	MPa
AF recovery rate	γ	80	–
Critical energy release rate	G_c	47.5	N/mm
Regularisation length	l_0	0.4	mm
Fatigue threshold	α_T	5 000	N/mm ²

4.4. Elastoplastic fatigue

The most physically complete regime combines J_2 plasticity with the fatigue phase field model. Two benchmarks are presented: a compact tension (CT) specimen made from GH4169 nickel-base superalloy at 550 °C, providing a quantitative comparison with experimental fatigue crack growth data, and a SENT specimen with power-law hardening, demonstrating the interaction between cyclic plastic straining and fatigue damage in a widely used reference geometry.

4.4.1. CT specimen (GH4169 superalloy, 550 °C)

The CT specimen benchmark follows the experimental and numerical study of Xie et al. [28] on GH4169 at elevated temperature. The specimen has width $W = 50$ mm and initial crack length $a_0 = 14$ mm. Cyclic loading is applied at load ratio $R = 0.1$ with maximum force $F_{\max} = 25$ kN. Material and phase field parameters are listed in Table 8. The plasticity model combines Voce isotropic hardening ($\kappa = \sigma_{y0} + Q_{\infty}(1 - e^{-b\alpha_p})$) with Armstrong–Frederick kinematic hardening, as calibrated for GH4169 at 550 °C in [28]. The fatigue threshold $\alpha_T = 5000$ N/mm² reflects the endurance limit of the superalloy; the fatigue driving force is based on the tensile elastic strain energy (mode I and mode II components) without a plastic-energy contribution, consistent with the formulation of [28]. Setting $\beta = 0$ for this benchmark reproduces the reference formulation: Xie et al. define the fatigue driving force purely from the tensile elastic strain energy, on the physical basis that at the load ratio $R = 0.1$ the incremental plastic stored energy per cycle is small compared with the elastic contribution. The plastic deformation nonetheless influences the fatigue response indirectly, by modifying the stress field through the Voce–Armstrong–Frederick hardening and thereby altering the elastic energy distribution ahead of the crack tip.

Fig. 13 presents the damage contours and the a – N curve. The damage band advances along the expected horizontal crack plane, driven by the tensile elastic strain energy (mode I and mode II components) which constitutes the crack-driving history variable $\mathcal{H}$. At each cycle, the plastic zone develops ahead of the crack tip and the cyclic plastic deformation feeds back into the stress field through the Voce–Armstrong–Frederick hardening, modifying the elastic energy distribution and indirectly accelerating fatigue-toughness degradation through the cumulative variable $\bar{\alpha}$. This coupling between cyclic plasticity and elastic driving force is physically essential for superalloys at elevated temperature, where the cyclic plastic deformation ahead of the crack tip is substantial even at the relatively high load ratio $R = 0.1$. The predicted a – N curve is in quantitative agreement with the experimental data, with a mean absolute error below 10% over the full 16 000-cycle history, and is consistently closer to the measured data than the reference numerical solution of Xie et al. [28]. The convergence of the phase field prediction towards the experimental data at long crack lengths is particularly notable, suggesting that the Voce–Armstrong–Frederick model with the chosen parameters correctly captures the crack-tip plastic zone evolution throughout the test.

Table 9

Material and phase field parameters for the SENT elastoplastic fatigue benchmark (Li et al. [29]). Power-law isotropic hardening: $\kappa(\alpha_p) = \sigma_{y0}(1 + E\alpha_p/\sigma_{y0})^n$.

Parameter	Symbol	Value	Unit
Young's modulus	E	210 000	MPa
Poisson's ratio	ν	0.3	–
Initial yield stress	σ_{y0}	440	MPa
Hardening exponent	n	0.1	–
Critical energy release rate	G_c	2.7	N/mm
Regularisation length	l_0	0.02	mm
Fatigue threshold	α_T	11.25	N/mm ²
Taylor–Quinney coefficient	β	1.0	–

4.4.2. SENT elastoplastic fatigue

The second elastoplastic fatigue benchmark reproduces the SENT specimen of Li et al. [29]. The geometry is identical to the elastic SENT of Section 4.1 (1×1 mm², $a_0 = 0.5$ mm) but now includes power-law isotropic hardening. Material and phase field parameters are listed in Table 9. Cyclic loading: $u_{\max} = 1 \times 10^{-3}$ mm, $R = 0$, a nominal upper limit of $N_{\text{total}} = 800$ cycles, and $\Delta N = 1$; the run is stopped earlier once the crack reaches the terminal length used in the reference comparison.

Figs. 14 and 15 present the damage, fatigue-degradation, plastic-strain, and von Mises stress contours together with the a – N curve. The plastic zone develops around the crack tip from the first cycle onwards, initially confined to a small region at the pre-crack tip but widening progressively as the crack advances and the crack-tip process zone sweeps through previously plastified material. The characteristic butterfly-shaped distribution of equivalent plastic strain, with lobes extending at $\pm 45^\circ$ to the crack plane, is consistent with the Mode I near-tip Prandtl slip-line field and has been widely observed in fatigue crack growth experiments. As the crack extends, fresh material ahead of the tip is cyclically loaded into the plastic range, and the accumulated plastic strain augments the fatigue variable $\bar{\alpha}$ through the $\beta = 1.0$ term, accelerating toughness degradation and sustaining crack propagation. The a – N curve matches the reference solution of Li et al. [29] to within 5% over the computed propagation history, which terminates before the nominal 800-cycle upper limit when the crack reaches the prescribed terminal length. The slight difference in crack growth rate in the mid-propagation regime ($N \approx 300$ – 600 cycles) is consistent with the known sensitivity of power-law hardening models to the exact near-tip plastic strain gradient, which in turn depends on the mesh resolution relative to the plastic zone size.

5. Computational performance

Mesh-convergence studies were conducted for all four regimes. In each case the normalised mesh size l_0/h is varied from 1 to 6 and the key output quantity (peak load for fracture benchmarks; crack length at a fixed cycle count for fatigue benchmarks) is recorded. A ratio $l_0/h \geq 2$ is sufficient for elastic problems; plastic fracture and fatigue problems require $l_0/h \geq 3$ due to the additional length-scale interaction between the phase field band and the plastic zone.

Fig. 16 substantiates this requirement quantitatively for the elastoplastic-fatigue SENT benchmark, with the phase-field length scale fixed at $l_0 = 0.020$ mm and the ratio l_0/h swept from 1 to 6. When only one element resolves l_0 (the case $l_0/h = 1$), the predicted final crack advance is approximately 29% below the converged baseline (under-resolved), as shown in Fig. 16(b) and (d). From $l_0/h = 2$ onwards, the relative error in the final crack advance stays below 1% for all subsequent refinements. The wall-clock cost in Fig. 16(c), however, grows approximately quadratically with l_0/h , from 14 min at $l_0/h = 1$ to 597 min at $l_0/h = 6$ (a 43× increase for a 6× refinement of l_0/h , consistent with a scaling of about $(l_0/h)^{2.1}$). As a balance between accuracy and efficiency, $l_0/h \geq 3$ is adopted in the production runs reported below, which gives an error of about 0.4% at a moderate cost of approximately 139 min.

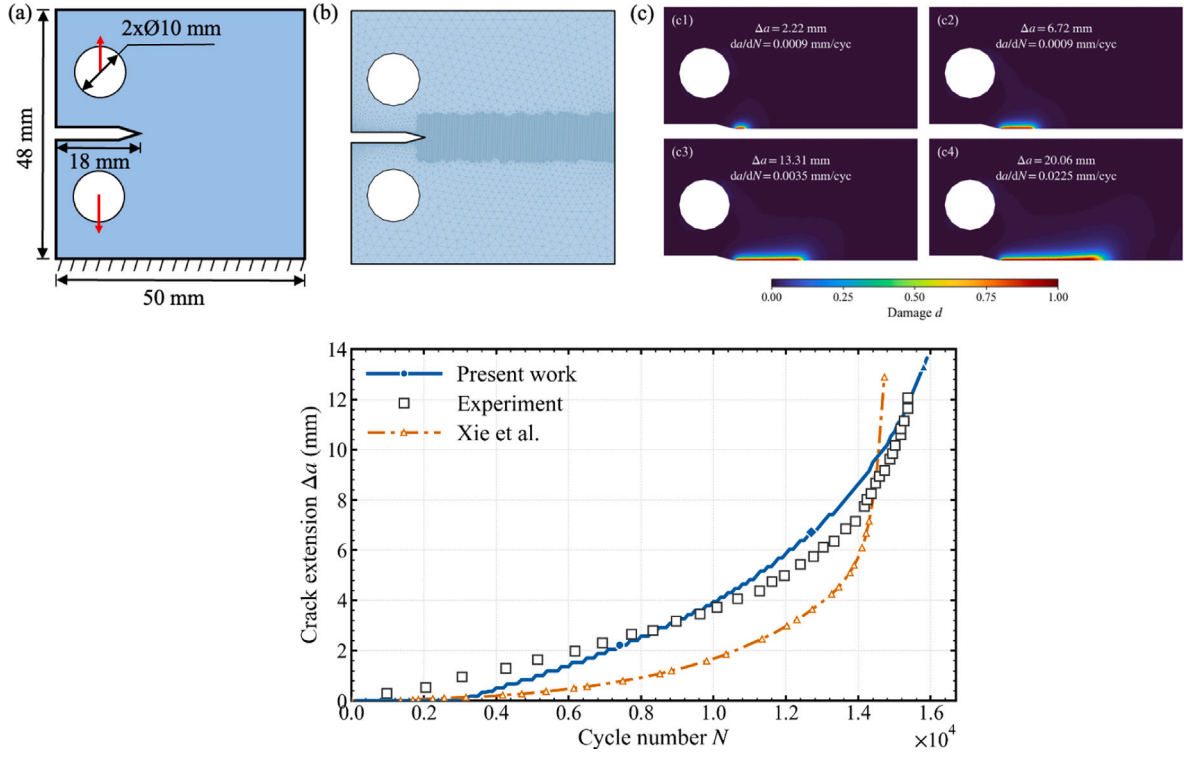


Fig. 13. CT specimen elastoplastic fatigue (GH4169, 550 °C, Xie et al. [28]): (top row) phase field damage d contours at $N = 4000$, 8000, 12000, and 16000 cycles showing the crack advancing along the horizontal plane; (bottom) predicted a - N curve compared with the numerical solution of Xie et al. [28] and experimental data. The present prediction lies closer to the experimental data than the reference numerical solution, with a mean absolute error below 10% over the full loading history.

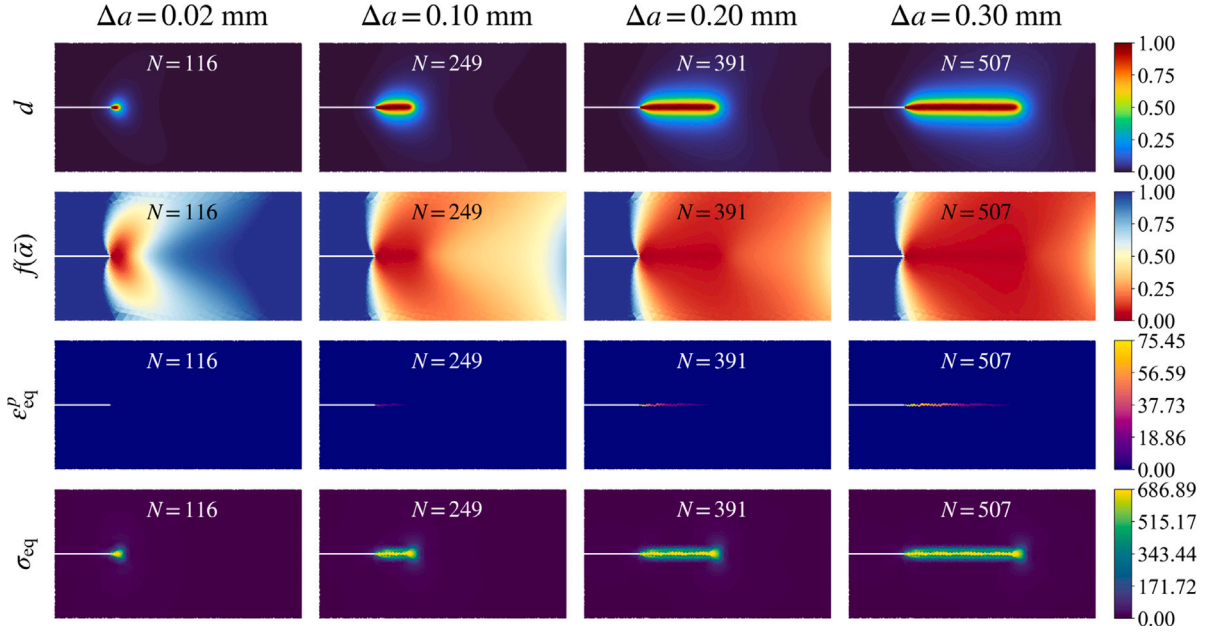


Fig. 14. SENT elastoplastic fatigue (Li et al. [29] Fig. 9): phase field damage d , fatigue-degradation function $f(\tilde{a})$, equivalent plastic strain ϵ_{eq}^p , and von Mises equivalent stress σ_{eq} at four cycle counts, arranged from top to bottom. The crack grows from the pre-crack tip while the butterfly-shaped plastic zone surrounding the advancing damage band widens with propagation.

The sensitivity of the predictions to the phase-field length scale l_0 itself, at a fixed mesh resolution, is reported in Fig. 17. Three values $l_0 \in \{0.012, 0.015, 0.018\}$ mm are scanned for the elastic SENT benchmark in panels (a)–(b), and five values $l_0 \in \{0.016, 0.018, 0.020, 0.022, 0.024\}$ mm for the elastoplastic-fatigue SENT in panel (c). For the elastic case, the load–displacement curves are visually indistinguishable, and the peak

force F_{peak} deviates from the mean by at most 1.13%, with the smallest length scale ($l_0 = 0.012$ mm) marginally above the $\pm 1\%$ band; the two larger values are within the band. For the elastoplastic-fatigue case, the a - N curves run nearly parallel and the final crack advance varies by less than 0.7% across the five values. Panel (d) reports the wall-clock cost of the two studies side by side: the elastic runs require 12–30 min,

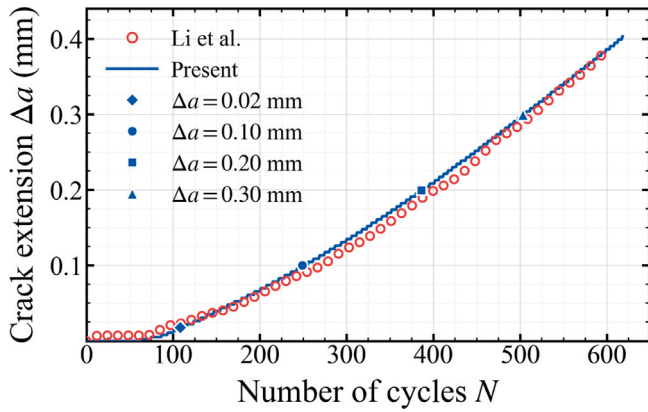


Fig. 15. SENT elastoplastic fatigue: predicted crack length a vs cycle number N compared with the reference solution of Li et al. [29]. The present prediction matches the reference closely until the terminal crack length is reached; the slight difference in growth rate in the mid-propagation regime reflects the sensitivity of power-law hardening to the exact near-tip plastic strain distribution.

roughly an order of magnitude less than the corresponding fatigue runs at comparable l_0 (83–191 min), and both groups become faster as l_0 grows.

Fig. 18 summarises the wall-clock time required to complete each of the eight benchmark cases on a desktop workstation (Intel i9-14900K, 64 GB RAM, Ubuntu Linux), presented as a horizontal bar chart grouped by physical regime. Elastic fracture cases are the least expensive, completing in 20–171 min depending on problem size and the number of load steps. Ductile fracture cases require 29–69 min; the relatively wide range reflects the additional cost of return-mapping iterations and the varying levels of plasticity across specimens. Elastic fatigue cases (17–54 min) are comparable in cost to ductile fracture, because the fatigue degradation function adds only modest overhead per cycle while the cycle-jump scheme keeps the number of fully resolved increments low. The two elastoplastic fatigue cases are by far the most expensive (195–295 min), reflecting the combined cost of return mapping, history-variable updates, and the large total number of representative cycle blocks required to capture crack growth over thousands of real cycles.

To characterise the cost structure of the elastoplastic fatigue simulations in more detail, Fig. 19 presents the timing breakdown for the SENT elastoplastic fatigue benchmark ($\Delta N = 1$, meaning every real fatigue cycle is fully resolved). Panel (a) shows the crack length a and cumulative wall time as functions of the cycle number N . The cumulative time grows nearly linearly with N , confirming that the per-cycle cost is approximately constant throughout the simulation; the total cost of approximately 295 min for 620 fully resolved cycles is consistent with the benchmark-level figure in Fig. 18. Crack initiation occurs at $N \approx 94$; before initiation, the specimen accumulates fatigue damage uniformly and a remains at its initial notch length. After initiation, the crack propagates steadily. Panel (b) shows the per-cycle wall time as a function of N , coloured by instantaneous crack length. The per-cycle cost is approximately 25–37 s, with moderate scatter attributable to variable plasticity-driven Newton–Raphson convergence behaviour across the load cycle. No systematic dependence on crack length is observed, confirming that the dominant cost is the plastic return mapping rather than the growing number of damaged elements.

The benefit of cycle-jump acceleration is quantified in Fig. 20 for the CT GH4169 benchmark ($\Delta N = 100$). Panel (a) shows the crack advance Δa and the cumulative wall time as a function of the real cycle number N . The crack advance remains nearly zero during the early incubation stage and then accelerates as fatigue-induced material degradation reaches the critical threshold; the cumulative wall time

grows monotonically, while the slope (cost per cycle block) stays modest throughout. The simulation covers approximately 16,000 cycles and completes in 195 min. Panel (b) shows the equivalent per-real-cycle cost as a scatter plot coloured by crack advance, together with the key efficiency metrics. Running the same simulation without cycle-jump acceleration ($\Delta N = 1$) would require an extrapolated total of approximately 18,964 min, obtained by multiplying the per-cycle cost measured at $\Delta N = 100$ by the full number of real cycles. This is approximately 100× longer than the actual 195 min. The extrapolated estimate assumes constant per-cycle cost and has not been verified by a direct $\Delta N = 1$ run; it serves as a conservative lower bound on the speedup, which is consistent with the theoretical ΔN -fold prediction.

The single-step result in Fig. 20 ($\Delta N = 100$, 195 min) demonstrates that cycle-jump acceleration is effective, but does not by itself indicate whether $\Delta N = 100$ is the optimal choice or whether a larger ΔN would noticeably degrade the predicted fatigue life. To address this, Fig. 21 reports a sweep of the cycle-jump step size from $\Delta N = 100$ to $\Delta N = 300$ on the same CT GH4169 benchmark. The crack-length histories in panel (a) follow a consistent shape and only differ in the cycle count at which the failure threshold $a = 36$ mm is reached. The interpolated fatigue lives N_{failure} in panel (b) span the range from approximately −12% to +9% relative to the five-case mean ($\approx 19,100$ cycles); the largest deviation of about 12% occurs at $\Delta N = 100$ (under-prediction), while $\Delta N = 300$ over-predicts the mean by about 9%. The wall-clock cost in panel (c) decreases monotonically with ΔN , from 195 min at $\Delta N = 100$ to 85 min at $\Delta N = 300$, while the number of phase-field/equilibrium evaluations in panel (d) drops correspondingly from 168 to 70. Taken together, $\Delta N \in [150, 250]$ is identified as a balanced operating range in which the predicted fatigue life is within about $\pm 6\%$ of that obtained at $\Delta N = 100$ while the cost is reduced by 25%–50%.

6. Conclusions

This paper has presented an open-source Julia/Gridap library for phase-field modelling of fracture and fatigue in elastic and elastoplastic solids. The library integrates four physical regimes (brittle fracture, elastic fatigue, ductile fracture, and elastoplastic fatigue) within a layered architecture, with the aim of supporting reproducible studies and cross-regime comparisons. The principal contributions are summarised below.

- To the best of our knowledge, the library is one of the first open-source implementations that combines elastic fracture, elastic fatigue, ductile fracture, and elastoplastic fatigue within the same codebase. The four regimes share a common variational skeleton and solver structure, so additional physical complexity is introduced by activating extra constitutive ingredients rather than by changing the solver itself.
- Using Gridap’s weak-form API, the governing variational equations are written closely to their analytical form. The single-language Julia environment reduces the gap between quadrature-point constitutive updates and global finite-element assembly, which makes the implementation easier to inspect and modify.
- Eight benchmark problems are reported, ranging from single-edge-notched brittle fracture to GH4169 nickel-base superalloy compact-tension elastoplastic fatigue. The predicted load-displacement curves and crack-growth histories are in good agreement with published numerical and experimental results. Cycle-jump acceleration makes high-cycle elastoplastic fatigue simulations feasible on a workstation.
- The package environment can be reproduced with a single command, and all benchmark scripts are self-contained and do not require pre-generated meshes. An optional real-time graphical interface is provided for interactive case inspection. The complete source code is openly available to support reproducible studies.

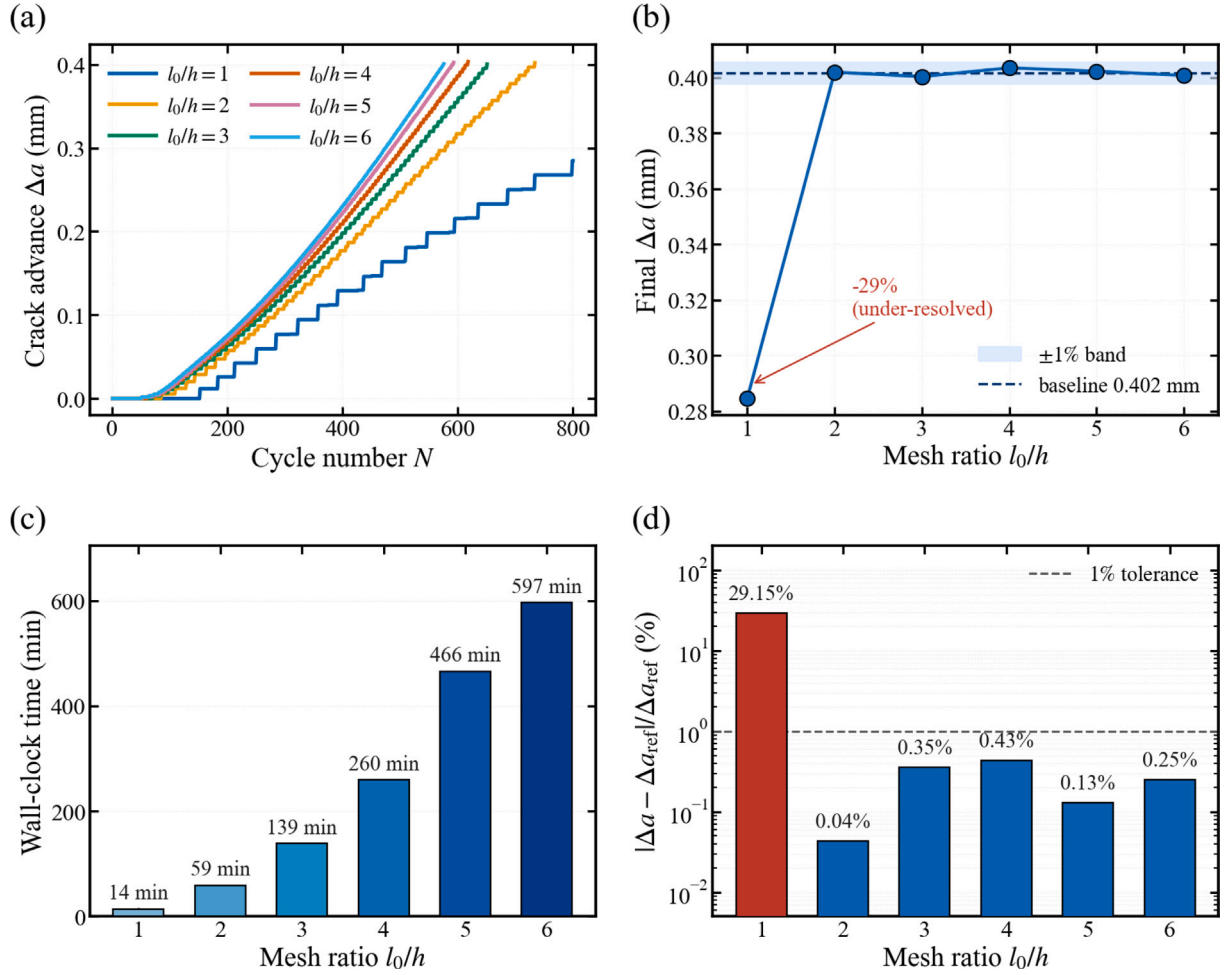


Fig. 16. Mesh-convergence study on the elastoplastic-fatigue SENT benchmark with $l_0 = 0.020$ mm. (a) Crack-advance histories Δa - N for the six mesh ratios $l_0/h \in \{1, \dots, 6\}$. (b) Final crack advance Δa as a function of l_0/h , with the light-blue band marking the converged $\pm 1\%$ range. (c) Total wall-clock time for each case. (d) Relative error in the final Δa with respect to the converged baseline ($l_0/h \geq 2$ mean), shown on a logarithmic axis with the grey dashed line marking the 1% tolerance.

The library is currently restricted to two-dimensional, small-strain, quasi-static problems, with a serial solver and a Gauss-Seidel staggered scheme. Three-dimensional simulations, finite-strain plasticity, parallel and adaptive solvers, more advanced cycle-jump strategies, and cohesive phase-field formulations such as the unified PF-CZM framework [50] are not yet supported and remain open directions for future work.

The unified architecture and the benchmark dataset also provide a basis for closer integration between data-driven methods and phase-field simulation. The load-displacement curves, damage-field evolutions, and crack-growth histories generated consistently across the four regimes can serve as training data for surrogate modelling. Recent studies on data-driven fatigue prediction have shown that neural-network surrogates can extract robust trends from limited datasets [52], and transfer-learning strategies further improve data efficiency across materials and loading conditions [53]. Built on datasets of this kind, physics-informed and energy-based surrogate models may help reduce the cost of high-cycle fatigue and complex-geometry fracture simulations, and may also be coupled with the present solver for parameter identification and uncertainty quantification. The layered architecture provides a starting point for extension towards three-dimensional fracture, thermo-elastoplastic coupling, and component-scale fatigue prediction.

CRediT authorship contribution statement

Peng Zhang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dazhi Zhao:** Visualization, Validation, Software. **Keke Tang:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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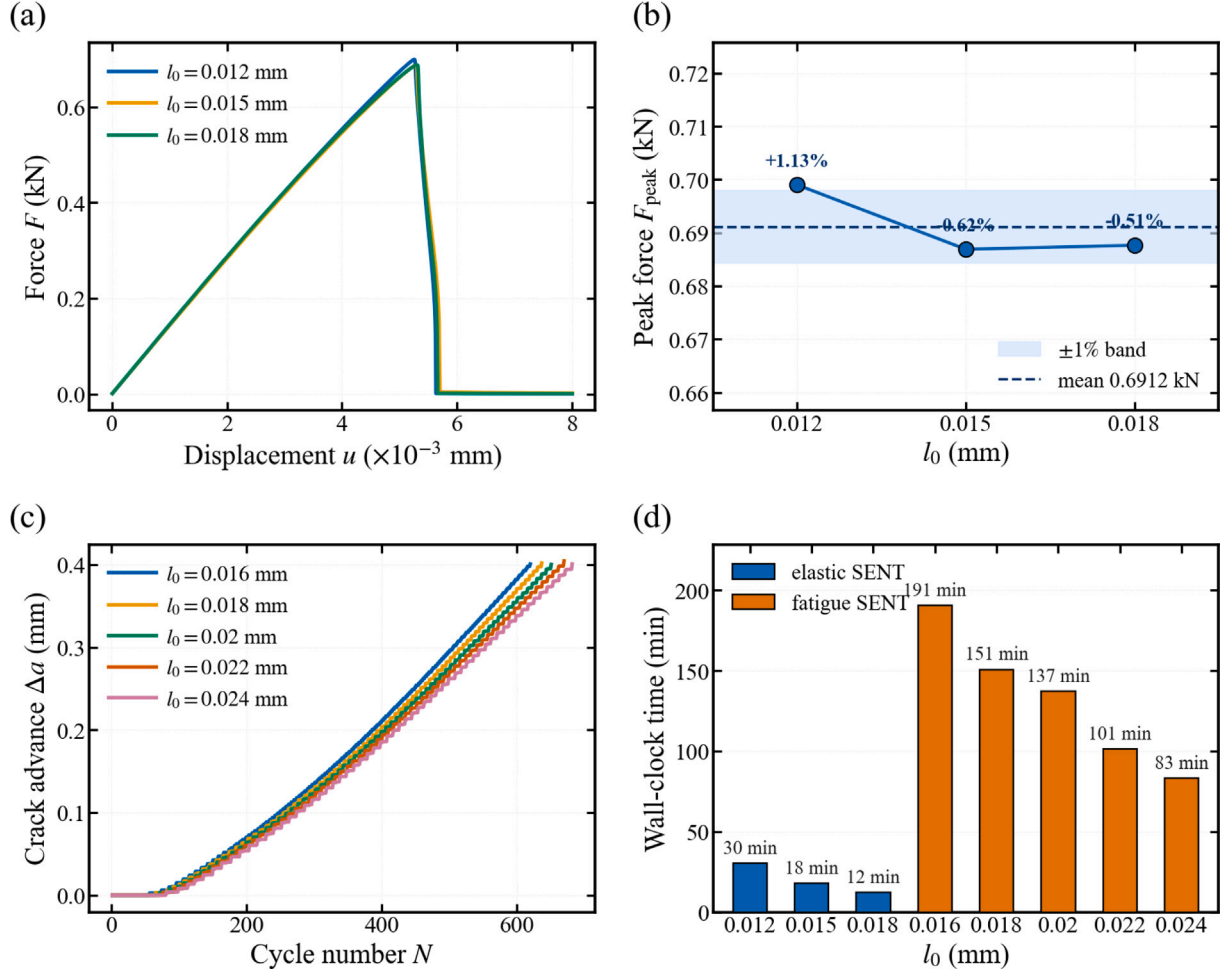


Fig. 17. Sensitivity to the phase-field length scale l_0 on the SENT benchmark. (a) Force–displacement curves of the elastic SENT for $l_0 \in \{0.012, 0.015, 0.018\}$ mm. (b) Peak force F_{peak} vs l_0 , with the light-blue band marking the $\pm 1\%$ range around the mean and percentage deviations annotated next to each marker. (c) Crack-advance histories Δa – N of the elastoplastic-fatigue SENT for $l_0 \in \{0.016, \dots, 0.024\}$ mm. (d) Wall-clock cost of the two studies, blue bars for the elastic runs and orange bars for the fatigue runs.

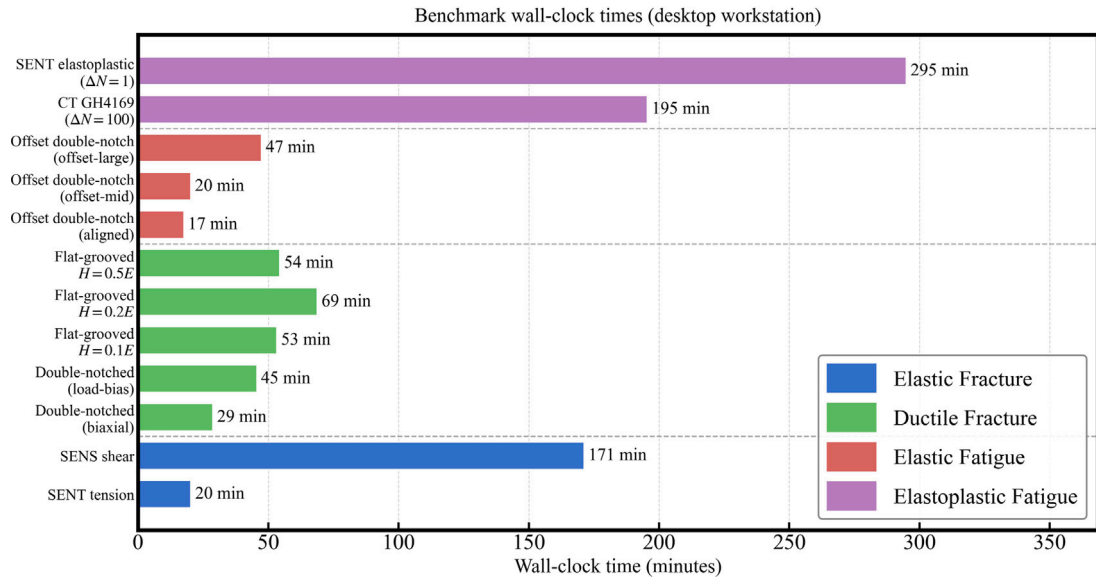


Fig. 18. Wall-clock times for all eight benchmark cases on a desktop workstation (Intel i9-14900K, 64 GB RAM, Ubuntu Linux). Bars are coloured by physical regime: elastic fracture (blue), ductile fracture (green), elastic fatigue (red), and elastoplastic fatigue (purple). Dashed horizontal lines separate the four groups.

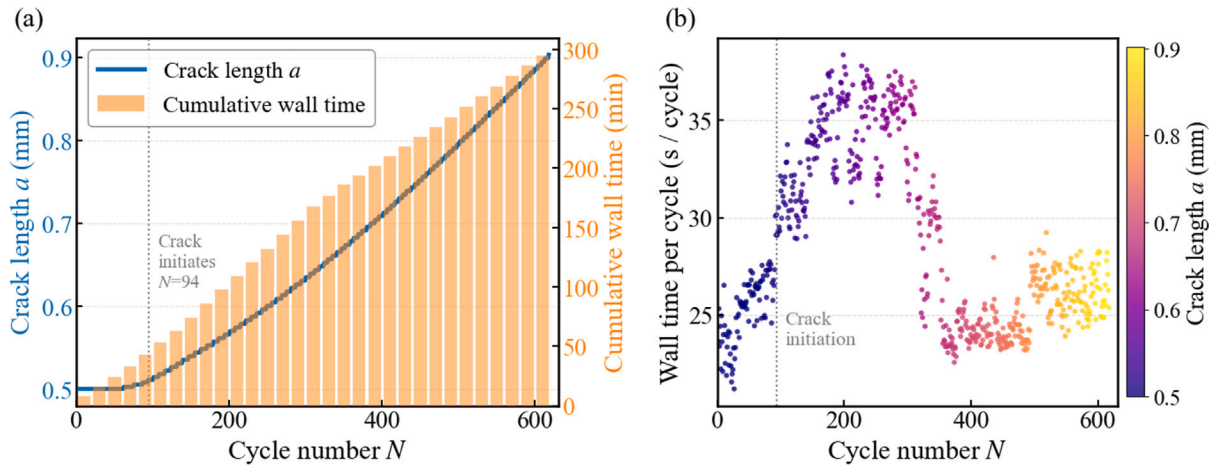


Fig. 19. Computational cost of the SENT elastoplastic fatigue benchmark ($\Delta N = 1$, 620 fully resolved cycles). (a) Crack length a (blue line, left axis) and cumulative wall time (orange bars, right axis) as functions of cycle number N . Each bar height equals the cumulative simulation time up to that cycle, forming a rising ramp that quantifies total elapsed cost. The dotted vertical line marks crack initiation at $N \approx 94$. (b) Per-cycle wall time as a function of N , with scatter points coloured by crack length a . The average per-cycle cost of approximately 28 s is broadly independent of crack length, indicating that the dominant computational overhead originates from the elastoplastic constitutive update rather than the increasing damaged-zone volume.

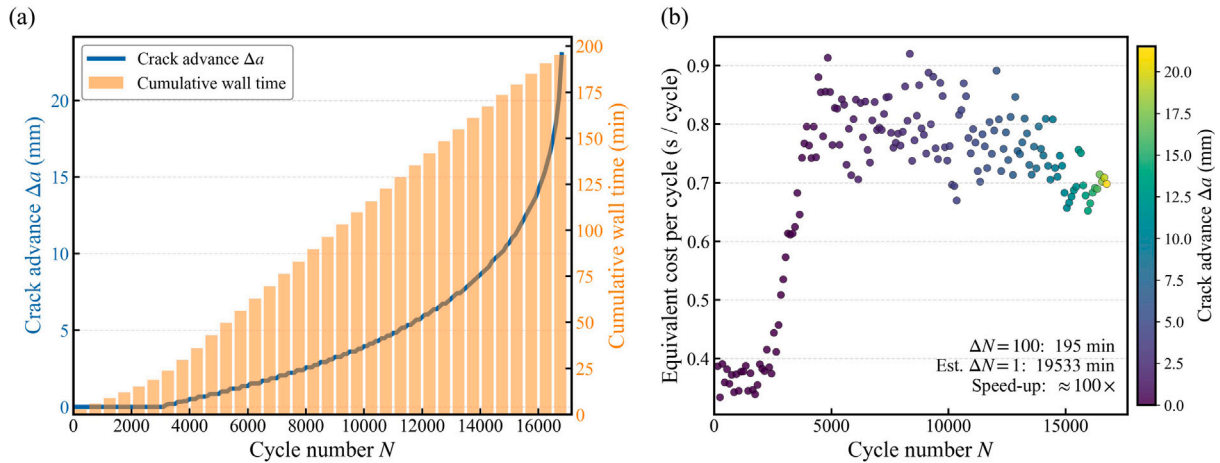


Fig. 20. Computational cost and cycle-jump efficiency for the CT GH4169 elastoplastic fatigue benchmark ($\Delta N = 100$, 16,000 cycles). (a) Crack advance Δa (blue line, left axis) and cumulative wall time (orange bars, right axis) as functions of the real cycle number N . Each bar represents the cumulative simulation cost at the end of a 500-cycle bin. (b) Equivalent per-real-cycle wall time, coloured by crack advance. The inset shows the actual simulation cost at $\Delta N = 100$ (195 min) and the extrapolated cost for a hypothetical $\Delta N = 1$ run (≈ 18,964 min, obtained by scaling the measured per-cycle cost by the total number of real cycles), yielding an estimated speedup of approximately 100x.

Appendix A. Code structure and usage guide

A.1. Why Julia?

Julia is a high-level, dynamically typed language compiled just-in-time to native machine code that runs at speeds competitive with Fortran and C++. For phase-field solvers, two properties are particularly relevant. First, its syntax is close to mathematical notation: variational residual forms, energy densities, and tangent operators are expressed with standard tensor arithmetic that maps closely onto the equations in Section 2, without a bridging layer between the mathematical specification and the runnable code. Second, Julia provides a single-language environment: return-mapping algorithms at the quadrature-point level, staggered solver logic at the global level, post-processing, and testing all live in the same package ecosystem, with uniform debugging, profiling (@profile, BenchmarkTools.jl), and unit-testing (Test.jl) tooling. There is no C++/Python boundary to cross when a constitutive model needs to be modified.

For context, the dominant alternatives for phase field implementation each carry a specific trade-off. ABAQUS UEL/UMAT subroutines benefit from mature industrial solvers but cannot be freely distributed and require repetitive boilerplate for every new constitutive feature. FEniCS provides a strong weak-form interface but its C++/Python two-language structure makes quadrature-point debugging cumbersome. MOOSE is designed for massively parallel production runs and has a deep inheritance hierarchy that is well-suited to large teams but adds overhead for single-investigator prototyping. Julia + Gridap occupies a different point in this space: fully open, single-language, with solver performance sufficient for all benchmark cases in this paper on a single desktop workstation.

A.2. Installation

Julia 1.9 or later is required. The recommended installation method is **juliaup**, the official cross-platform version manager:

```
1 # Linux / macOS
2 curl -fsSL https://install.julialang.org | sh
```

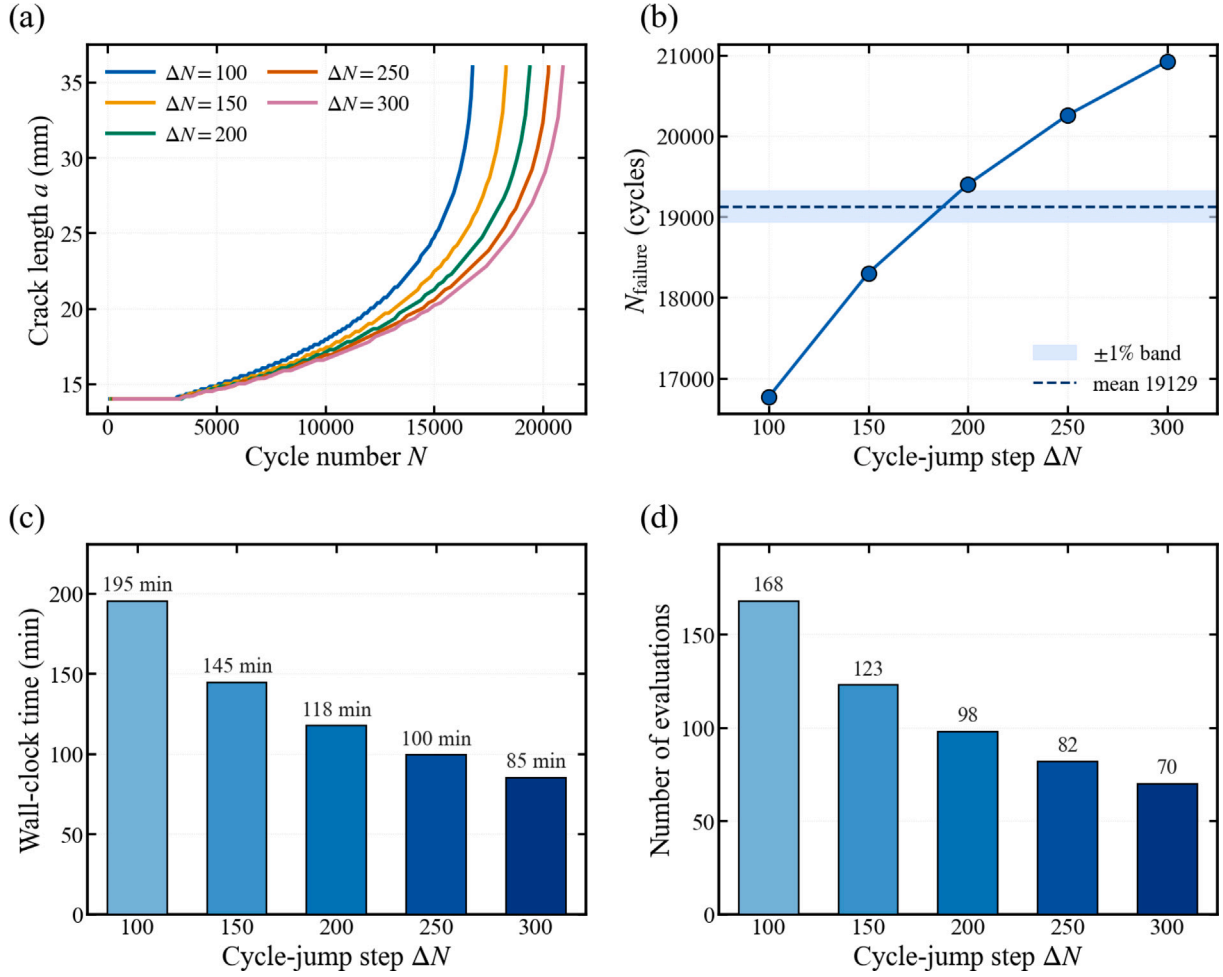


Fig. 21. Cycle-jump step-size sweep on the CT GH4169 elastoplastic-fatigue benchmark for $\Delta N \in \{100, 150, 200, 250, 300\}$. (a) Crack-length histories $a(N)$ for the five jumps, all linearly truncated at the failure crack length $a = 36$ mm. (b) Predicted fatigue life N_{failure} obtained by linear interpolation of the $a-N$ curves at $a = 36$ mm, with the light-blue band marking the $\pm 1\%$ range around the mean. (c) Wall-clock time. (d) Number of phase-field/equilibrium evaluations.

```
3 # Windows (PowerShell)
4 winget install julia -s msstore
```

After installing Julia, clone the repository and instantiate the environment:

```
1 git clone https://github.com/FCMzp05/
  PhaseFieldFracture.jl
2 cd PhaseFieldFracture.jl
3 julia --project=. -e "using Pkg; Pkg.instantiate()"
```

This downloads and pins all dependencies to their declared versions, ensuring full computational reproducibility across platforms and Julia upgrades. **Note:** the first execution of any script incurs a one-time JIT compilation overhead (typically 30–90 s) while Julia compiles the Gridap kernels to native code; all subsequent runs of the same Julia session are significantly faster. The key packages and their roles are listed in Table A.1.

A.3. Repository structure

The benchmark scripts are organised into four subdirectories corresponding to the four physical regimes:

```
1 BenchmarkCases_Organized/
2 ElasticFracture/
3 ElasticFracture_SENT_Tension.jl # Sec. 4.1 -- AT2
  , Miehe spectral + Amor
```

```
4 ElasticFracture_SENS_Shear.jl # Sec. 4.1 -- AT2,
  mixed-mode shear crack
5 PlasticFracture/
6 PlasticFracture_DoubleNotched.jl # Sec. 4.2 -- J2
  plasticity, linear hardening
7 PlasticFracture_FlatGrooved.jl # Sec. 4.2 --
  offset notched plate, H scan
8 ElasticFatigue/
9 ElasticFatigue_TPB_AT1.jl # Sec. 4.3 -- AT1
  three-point bending, brittle
10 ElasticFatigue_OffsetDoubleNotch.jl # Sec. 4.3 --
  offset double-notch, DeltaN=1, save every 20
11 PlasticFatigue/
12 PlasticFatigue_SENT.jl # Sec. 4.4 -- SENT, J2
  + power-law, DeltaN=1
13 PlasticFatigue_CT.jl # Sec. 4.4 -- CT,
  Chaboche (Voce+A-F), DeltaN=100
```

Each script is self-contained: geometry is defined analytically through Gmsh's scripting API and meshed at runtime; the simulation runs without any pre-generated mesh file; and all output (VTU snapshots, CSV force-displacement or crack-growth data) is written to a case-named subdirectory. A brief description of each script follows.

- **ElasticFracture_SENT_Tension.jl:** 1×1 mm² square plate with a horizontal notch at mid-height. Implements AT2 with both Miehe spectral and Amor volumetric-deviatoric energy splits; the two results are compared.

Table A.1
Key Julia packages used in the library.

Package	Role
Gridap.jl	Finite element assembly, function spaces, weak-form DSL, sparse linear algebra
GridapGmsh.jl	Interface between Gmsh mesh files (.msh) and Gridap triangulations
Gmsh.jl	Parametric mesh generation and geometry scripting via the Gmsh API
GLMakie.jl	Real-time interactive visualisation for the graphical user interface
Plots.jl	Static post-processing plots (load–displacement curves, a - N histories)
WriteVTK.jl	VTK/VTU output for ParaView post-processing
LinearAlgebra.jl	Dense and sparse matrix operations (part of Julia standard library)
DelimitedFiles.jl	CSV output for load–displacement and crack-growth data

- **ElasticFracture_SENS_Shear.jl**: Same geometry loaded in shear (horizontal displacement imposed on the top edge). Demonstrates mixed-mode crack kinking under predominantly Mode II loading.
- **PlasticFracture_DoubleNotched.jl**: Double-notched rectangular specimen under tension. Employs J_2 plasticity with linear isotropic hardening ($H = 300$ MPa) in the effective stress space and the AT2 model with the Amor split.
- **PlasticFracture_FlatGrooved.jl**: Rectangular plate with two asymmetrically offset semicircular notches under plane-strain tension. Scans the linear hardening modulus $H/E \in \{0.1, 0.2, 0.5\}$ to quantify the sensitivity of predicted fracture strain to the hardening level.
- **ElasticFatigue_TPB_AT1.jl**: Three-point bending specimen with a central notch. Uses the AT1 model (with elastic threshold) and the threshold-based fatigue degradation framework. Crack-growth rate and damage evolution are studied over 400 load cycles.
- **ElasticFatigue_OffsetDoubleNotch.jl**: Double-notched panel with offset notches under cyclic tension-compression. Demonstrates crack initiation from both notches and subsequent coalescence. VTU snapshots are saved every $\Delta N_{\text{save}} = 20$ representative cycles up to 400 cycles.
- **PlasticFatigue_SENT.jl**: Single-edge-notched tension specimen with J_2 plasticity (power-law hardening) and cycle-jump stride $\Delta N = 1$. Every real fatigue cycle is fully resolved. Demonstrates the classic butterfly-shaped plastic zone accompanying fatigue crack growth.
- **PlasticFatigue_CT.jl**: Compact tension specimen loaded at $R = 0.1$ using Chaboche plasticity (Voce isotropic + Armstrong–Frederick kinematic hardening) and $\Delta N = 100$. Fatigue driving force is purely elastic (Taylor–Quinney weight $\beta = 0$). Models fatigue crack growth in a nickel-base superalloy compact-tension specimen under cyclic loading.

A.4. Parameter layout of a benchmark script

Every script groups its parameters into three blocks. The example below is taken from PlasticFatigue_CT.jl (CT specimen, GH4169 superalloy, Section 4.4):

```

1 # --- Geometry ---
2 const W_CT = 50.0      # specimen width (mm)
3 const H_CT = 48.0      # specimen height (mm)
4 const V_TIP_X = 14.0   # initial crack length a0 (mm)
5 const PIN_RADIUS = 5.0 # loading pin radius (mm)
6
7 # --- Material: GH4169 (elastic + Voce isotropic + A-F
8   kinematic) ---
9 const E = 161_000.0    # Young's modulus (MPa)
10 const nu = 0.3         # Poisson's ratio
11 const sigma_y0 = 900.0 # initial yield stress (MPa)
12
13 const Q_inf = 80.0     # Voce saturation stress (MPa)
14 const b_voce = 5.0     # Voce saturation rate

```

Table A.2
VTK output fields written by all simulation scripts.

Field name	Symbol	Definition
d	d	Phase field damage variable
u	$\mathbf{u}$	Displacement field
epspeq	α_p	Equivalent plastic strain
psip	ψ^p	Plastic stored energy density
sigmaeq	$\bar{\sigma}_{eq}$	Von Mises equivalent stress
alpha_bar	$\bar{\alpha}$	Cumulative fatigue history variable
falpha	$f(\bar{\alpha})$	Fatigue degradation function

```

13 const C_AF = 12_000.0 # A-F kinematic modulus (MPa)
14 const gamma_AF = 80.0 # A-F recall parameter
15
16 # --- Phase field + fatigue ---
17 const Gc = 47.5 # critical energy release rate (N/mm)
18 const l0 = 0.4 # regularisation length (mm)
19 const alpha_T = 5000.0 # fatigue threshold
20 const beta_TQ = 0.0 # Taylor-Quinney weight (elastic driving only)
21 const CYCLE_JUMP = 100 # cycle-jump stride DeltaN
22 const N_TOTAL = 60_000 # total real cycles

```

For elastic fracture scripts the plasticity block is absent; for monotonic (non-fatigue) scripts the fatigue block is absent. All other structural elements (mesh generation, FE space definition, staggered solver loop, and output) are identical across regimes, reflecting the layered design of the library.

A.5. Running a simulation

A benchmark is executed from the repository root as:

```

1 julia --project=. BenchmarkCases_Organized/
  PlasticFatigue/PlasticFatigue_CT.jl

```

Outputs are written to a subdirectory named after the case. VTU snapshot files contain the fields listed in Table A.2 and can be opened directly in ParaView (<https://www.paraview.org>); applying the *Warp by Vector* filter to the $\mathbf{u}$ field provides a convenient view of deformation superimposed on damage or plastic-strain contours. Load–displacement curves and crack-growth histories are written as plain CSV files for downstream plotting with any tool.

A.6. Extending the library

Adding a new benchmark to the library requires three steps. First, create a new .jl script in the appropriate regime subdirectory and copy the parameter header from an existing script in the same regime (or the next higher regime if new physics are needed). Second, implement any new constitutive update inside a `constitutive_update!` function that follows the signature of the existing functions; the staggered solver loop calls this function without modification. Third, register the new case in the README.md table. No changes

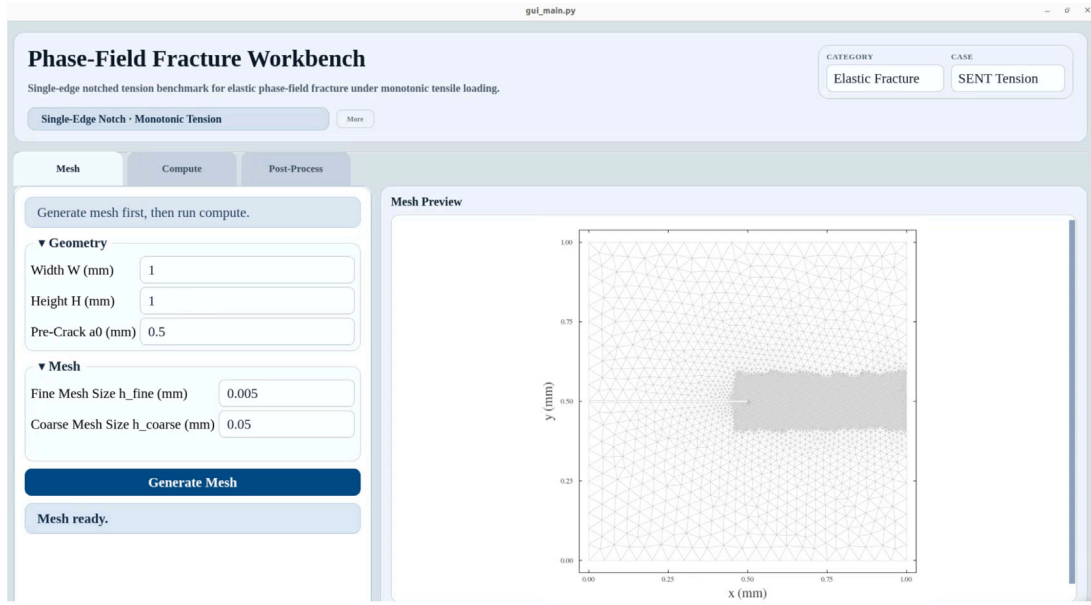


Fig. B.1. Demonstration GUI: *Mesh* page. The screenshot shows the case header with category and case selectors, together with the geometry and discretisation input panel used to generate a benchmark mesh before solving. In the current GUI code, this page collects editable geometry and mesh fields from the case registry and displays a mesh preview so that users can verify the benchmark setup visually before launching the computation.

to any shared infrastructure are necessary. To add a new physical regime (e.g. thermo-mechanical coupling), the shared solver skeleton can be extended by adding temperature as an additional finite-element field and supplying the corresponding residual forms; Gridap's multi-field MultiFieldFESpace abstraction handles the assembly automatically.

The dispatch-based design also accommodates phase-field formulations beyond AT1 and AT2. As a representative example, cohesive phase-field models [50] can be added with the following modifications: (i) a new dispatch tag (e.g. `::Val{:CZM}`) in `phasefield.jl`, providing the geometric crack function $w(d) = 2d - d^2$ and the corresponding energetic degradation function $g(d)$ together with the associated bilinear and linear forms; (ii) additional fields in `materials.jl` for the cohesive law parameters (material strength, shape parameters, characteristic length); and (iii) a wrapper in `solver.jl` that calls a Picard or Newton iteration for the phase-field subproblem in this case, since the cohesive degradation function is generally a nonlinear function of d . The remaining modules (`energy.jl`, `plasticity.jl`, the fatigue routines, and the outer staggered loop) are not affected. Implementing this extension is left for future work.

A.7. Reference implementation snippets

This subsection collects the reference Julia implementations of the kernel routines referenced from Section 3, namely the irreversibility clamp, the history-variable update, the radial return mapping, and the cycle-jump update. They are placed here so that the main text retains only the methodology and the high-level implementation choices, while the literal Julia code remains available for readers who wish to reproduce or modify the implementation. Variable names follow the mathematical notation of Section 3.

Irreversibility clamp. The pointwise clamp $d_{n+1} \leftarrow \max(d_{n+1}, d_n)$ used in Section 3.2 is implemented as

```
1 function enforce_irreversibility!(d_new, d_old)
2     for i in eachindex(d_new)
3         d_new[i] = max(d_new[i], d_old[i])
4     end
5 end
```

History-variable update. The crack-driving history variable update $H \leftarrow \max(H, \psi^{e+} + \beta\psi^p)$ used in Section 3.2 is realised as

```
1 function update_history!(H, psi_plus, psi_p, beta)
2     for i in eachindex(H)
3         H[i] = max(H[i], psi_plus[i] + beta*psi_p[i])
4     end
5 end
```

Radial return mapping. The full J_2 return-mapping algorithm of Section 3.3 is implemented at each quadrature point as

```
1 function return_mapping!(state, ee_trial, mat)
2     sig_tr = mat.C odot ee_trial # trial stress
3     sigma_eff_tr = sqrt(1.5*(sig_tr odot sig_tr)) # trial equivalent stress
4     s_tr = dev(sig_tr) # trial deviatoric stress
5     F_tr = seq_tr - sqrt(2/3)*kappa(state.alpha_p, mat)
6     if F_tr <= 0.0
7         state.sigma_eff = sig_tr; return # elastic step
8     end
9     # Newton iteration with linear-hardening closed form as initial guess
10    dg = F_tr / (3*mat.G + 2/3*kappa_prime(state.alpha_p, mat))
11    for k in 1:10
12        ap_new = state.alpha_p + sqrt(2/3)*dg
13        R = seq_tr - 3*mat.G*dg - sqrt(2/3)*kappa(ap_new, mat)
14        Rp = -3*mat.G - 2/3*kappa_prime(ap_new, mat)
15        dg -= R/Rp
16        abs(R) < 1e-12 && break
17    end
18    N_tr = 1.5*s_tr/seq_tr # flow direction
19    state.ep += dg*N_tr # update plastic strain
20    state.alpha_p += sqrt(2/3)*dg # update equivalent plastic strain
```

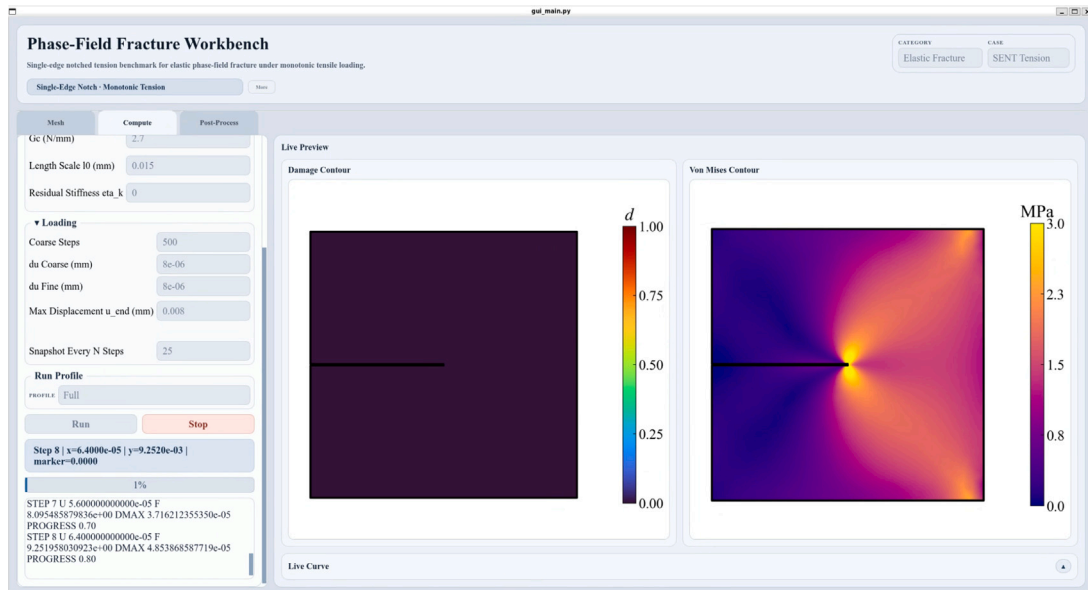


Fig. B.2. Demonstration GUI: *Compute* page. This page is used to launch the selected benchmark with a prescribed run profile and to monitor the current run through progress indicators and live preview panels. As implemented in the GUI code, the compute stage is intended to expose the solver workflow in an accessible way by showing representative contour updates during the run, rather than to provide a fully customisable industrial analysis dashboard.

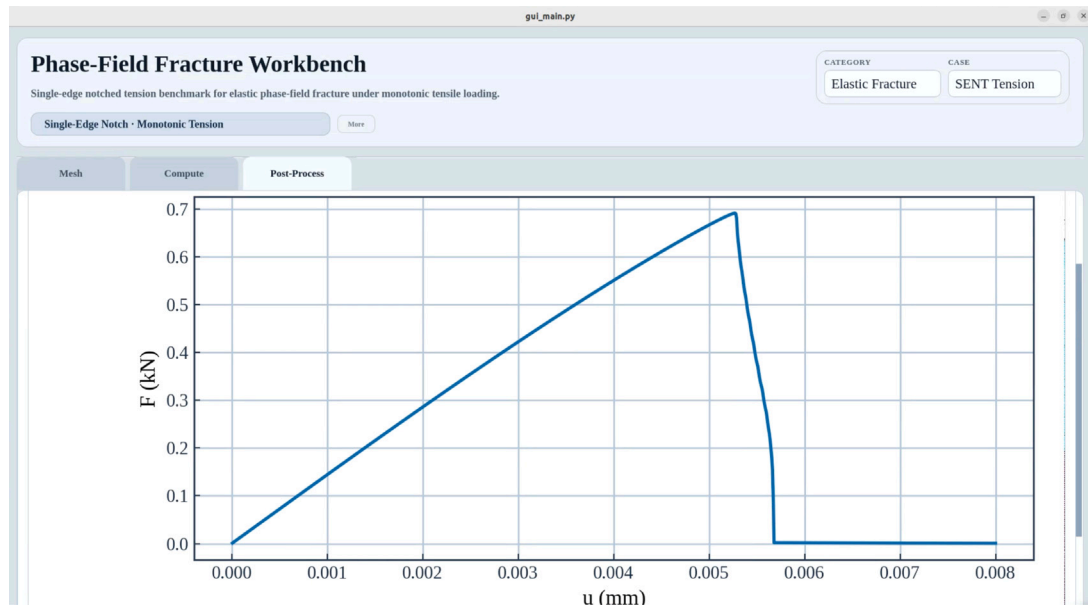


Fig. B.3. Demonstration GUI: *Post-process* page. After a run has completed, the interface displays selected output fields and curves for qualitative inspection of the benchmark result. In the present implementation, this page serves mainly as a compact visual summary of the final mesh, field contours, and response history, helping readers and students inspect benchmark outputs without manually opening the underlying result files.

```

21 state.sigma_eff = sig_tr - 2*mat.G*dg*N_tr #
    update effective stress
22 end

```

Cycle-jump update. The cycle-jump update of Section 3.4 advances the cumulative fatigue variable as follows. In this snippet, α_{tilde} denotes the value after completing one representative cycle, while $\alpha_{\text{tilde_old}}$ stores the value at the beginning of that representative cycle. If a representative cycle is discretised by multiple substeps, the caller first accumulates the positive substep contributions over the full cycle; the difference below is therefore the net positive increment of one representative cycle, not a single unloading-step difference.

```

1 function cycle_jump!(alpha_bar, alpha_tilde,
    alpha_tilde_old,
2     g_d, psi_plus, psi_p, beta, delta_N)
3 for i in eachindex(alpha_bar)
4     # alpha_tilde is evaluated after the
    representative cycle.
5     alpha_tilde[i] = g_d[i]*(psi_plus[i] + beta*
    psi_p[i])
6     # d_alpha represents the positive increment
    accumulated
7     # over one representative cycle before
    extrapolation.

```

```

8      d_alpha = max(0.0, alpha_tilde[i] -
9      alpha_tilde_old[i])
10     alpha_bar[i] += delta_N*d_alpha
11     alpha_tilde_old[i] = alpha_tilde[i]
12 end
end

```

Appendix B. Graphical user interface

A lightweight GUI prototype has been organised as a demonstration interface for the benchmark workflow. According to the current implementation in `gui_main.py` and `case_registry.py`, the top-level entry groups cases into four categories (*Elastic Fracture*, *Elastic Fatigue*, *Plastic Fracture*, and *Plastic Fatigue*) and exposes a guided three-page sequence: *Mesh*, *Compute*, and *Post-process*. The practical aim of this interface is to provide a unified visual entry point to the benchmark suite: users select a predefined case, inspect or modify a small set of geometry and discretisation parameters, launch a run through a restricted run-profile selector, and then review representative fields and curves without manually navigating the output directories. In this sense, the GUI serves mainly as a compact wrapper around the benchmark scripts and result manifests already discussed in Appendix A. It is intended primarily for teaching, presentation, and first-contact demonstration of the library, helping new users understand the mesh-generation, solution, and result-inspection workflow in a more accessible form. It should not be interpreted as a general-purpose finite element analysis environment, nor as a substitute for direct script-level control when reproducible research calculations are required. In practice, Linux is the recommended platform because the Python environment, Julia backend invocation, and GUI rendering have been tested most consistently there (see Figs. B.1–B.3).

Data availability

All the code has been made publicly available on GitHub; the link is <https://github.com/FCMzp05/PhaseFieldFracture>.

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